Surveillance and the Social Value of Information: Coordination Among LLM Agents

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Abstract

The value of information for collective action lies not only in its content but in its publicness: each person must be able to expect that others will read the same fact as a reason to move. I show that surveillance can destroy this public value without hiding any content, in a language-based Morris–Shin regime-change game played by large language model (LLM) agents who decide whether to join an uprising that succeeds only if enough of them join. Each agent reads a private natural-language briefing drawn from a noisy signal, so state and actionability cues arrive as prose, not as numbers. The agents reproduce the model's threshold comparative static: participation declines with regime strength (mean $r(J, A(\theta)) = +0.80$ across 7 models). Pre-play communication preserves this structure while moving participation only modestly ($\Delta = +2.10$ pp pooled, near zero equal-weighted across models). I then add surveillance only when peer messages are written, holding the receiver's decision prompt fixed. Participation falls at full scale in the primary model and in a nested 500-cell replication on a second model family ($\Delta = -8.0$ pp); four additional prompt-isolation reruns, of 20 matched cells each, add negative directional sign checks. A codedness control that makes messages *more* coded but says nothing about monitoring barely moves participation ($\Delta = -0.7$ pp), so generic vagueness is not the mechanism. The lost content is specific: surveilled senders keep the bad news but stop publicly claiming that the regime is beatable, and that judge-scored downgrade carries much of the action drop. Monitoring thus converts public information into merely private information—the publicness margin at the center of the social value of information—without hiding bad news or entering the receiver's choice. Because language models increasingly draft, filter, and relay human communication, these estimates directly characterize AI-mediated channels; whether they extend to unmediated human communication remains open.

Short abstract (100 words). The value of information for collective action depends on its publicness: each person must expect others to read the same fact as a reason to move. I show that surveillance can strip this public value without hiding the information, in a language-based Morris–Shin regime-change game with LLM citizens deciding whether to join an uprising that succeeds only if enough do. Seven models reproduce threshold-like signal use.

Sender-side surveillance cuts joining at full scale in two model families, while a codedness control that never mentions monitoring leaves joining nearly unchanged. Because language models increasingly carry human communication, the estimates characterize AI-mediated channels directly.

JEL: C72, C92, D82, D83, P16
Keywords: social value of information, global games, regime change, surveillance, communication, LLM agents, AI-mediated communication, coded speech

1 Introduction

Bad news alone does not produce collective action: people must also believe that others can read it as a reason to move. Economists have long recognized that in coordination settings the value of information depends on this second condition—on its *publicness*, not only its content (Morris and Shin, 2002). In a bank run (Diamond and Dybvig, 1983), a currency attack (Morris and Shin, 1998; Obstfeld, 1996), or a political uprising (Angeletos et al., 2007), a fact moves behavior only insofar as each actor expects others to act on the same fact. This paper asks whether that public value can be attacked directly: can an observer strip information of its coordinating power while leaving the information itself in circulation? I answer with a controlled experiment. Large language model (LLM) agents play a Morris–Shin regime-change game (Morris and Shin, 2003; Carlsson and van Damme, 1993; Frankel et al., 2003), exchange messages with trusted contacts, and decide whether to join an uprising that succeeds only if enough of them do. When peer messages are written under surveillance—and only the message stage changes, never the receiver's own decision prompt—joining falls sharply, even though the messages still report that the regime is weak. The mechanism turns on what a message does in a coordination crisis: it reports a state of the world, and it reveals the speaker's willingness to say so openly. Monitoring leaves the bad news in circulation while stripping its public, coordinating value—it converts public information into merely private information.

Two developments make this margin newly consequential. First, states that once censored now monitor. King et al. (2013) show that Chinese censorship tolerates even sharp criticism of the state while systematically deleting content with collective-action potential; Xu (2021) documents digital surveillance as preemptive control; Guriev and Treisman (2019) describe informational autocrats who rule through the management of information rather than mass violence. If monitoring the channel alone can strip information of its public value, a regime need not hide anything to be safe—an economical form of control that existing evidence cannot cleanly separate from ordinary deterrence. Second, a growing share of written communication is drafted, summarized, filtered, and relayed by language models. The message layer through which people coordinate is becoming, in part, an AI system—one whose output, this paper shows, responds systematically to cues about who is watching. Whether monitored AI channels preserve the public component of the information they carry is therefore not only a question about human behavior studied by proxy; it is a question about the channels themselves.

A regime has underlying strength that citizens observe only through noisy private signals.

Some see weakness; others see stability. Each citizen chooses whether to join an uprising or stay home, and joining pays only if enough others join. Staying home is safe but forgoes the chance of change. Because each citizen's best action depends on beliefs about others' actions and beliefs, coordination creates a higher-order regress. Carlsson and van Damme (1993) and Morris and Shin (2003) showed that sufficiently precise private signals collapse this regress to a unique threshold equilibrium: citizens join below the threshold and stay above it. The model therefore yields sharp comparative statics.

Laboratory experiments confirm threshold behavior with stylized numeric signals (Heinemann et al., 2004, 2009; Shurchkov, 2013; Szkup and Trevino, 2020; Helland et al., 2021), but do not capture the rich, multi-source information of real crises. Field evidence bundles strategic behavior with institutional differences, repression risk, and unobserved information structures. Surveillance adds another confound: observed declines in dissent combine personal deterrence with changes in peer communication. This gap persists for a practical reason. No field design can randomize surveillance of dissident communication while observing every message exchanged, and no human-subject experiment can hold the receiver's information environment fixed while varying only the incentives under which peers write. Nor can numeric-signal laboratory games represent the object surveillance actually distorts, which is language.

I isolate the second channel. The mechanism operates before action: monitoring changes what citizens say, and those messages change receivers' actions. Both effects appear in the experiment, while coded style alone does not reproduce the action shift. The receiver never sees a surveillance warning in the main treatment; only the sender's message-generation environment changes. Bad news remains in circulation but becomes less public and less usable as a shared reason to act.

I use LLM agents to measure a controlled language information structure. LLM agents are what make the ideal experiment feasible: every message is observed, the receiver's decision prompt is held bit-identical across arms, and surveillance is randomized on the sender side alone at scale—controls no field or human-subject design can deliver. The design then admits two readings. Narrowly, the estimand is a response function: how the hosted model snapshots in Table 2 (decoding configuration in Appendix F) respond to controlled changes in private prose signals and peer-message distributions. Broadly, because language models increasingly occupy the message layer itself, the same estimates characterize an AI-mediated communication channel directly rather than by proxy. I embed 7 models spanning 6 architecture families in the Morris and Shin (2003) game. Each agent receives a private signal $x_i = \theta + \varepsilon_i$ rendered as a natural-language intelligence briefing whose content shifts monotonically with signal strength across eight evidence domains. No explicit payoff table is provided; the stakes are embedded in the narrative, so agents must form action-relevant beliefs from language. I hold the latent state, private briefings, receiver's decision prompt, and decision rule fixed while changing only the incentives under which peer messages are produced. After verifying threshold-like signal use, I introduce surveillance.

The experiment first builds a language-mediated coordination environment disciplined by the canonical global game's comparative statics; Table 1 maps each canonical object to its implemented analogue. Participation decreases monotonically with regime strength and tracks the benchmark attack mass across 1,800 period-level observations and 7 models (mean $r = +0.80$, with $p < 0.001$ for every model), establishing threshold-like use of the prose signal.

Pre-play messages then carry regime-strength information and affect decisions, isolating a communication channel that field data cannot easily separate. The participation effect varies by model: the cells-pooled matched-cell estimate is +2.10 pp, but it is dominated by the primary model, and the equal-weighted across-model average is near zero (Appendix E, Table A7).

Surveillance changes only the incentives under which peers write messages. Joining falls in the primary model and in a fully nested 500-cell replication on a second model family, with the receiver's decision prompt held bit-identical. A codedness control elicits even more heavily coded messages without monitoring framing but leaves joining nearly unchanged, so generic coded form does not explain the surveillance effect. What agents report believing depends on what the belief question shows them: stated expectations stay stable when the belief prompt withholds treated peer messages (post-decision in the primary model and pre-decision in a nested replication), but move when it includes the messages available at the action stage.

Mechanism tests locate the treatment effect in peer messages. Message controls show that agents actually use peer messages when they decide. Bad news remains in circulation: in the primary model, explicit weakness references rise, while direct references to the regime and to action fall and metaphorical, deniable phrasing rises. A discriminant test in Llama 3.3 70B replaces the coordination decision with a private bet on regime fall, and surveillance still moves that task, establishing a downstream judgment shift outside the main JOIN/STAY action prompt. Because a bet on regime fall can still depend on expectations about others' participation, this contrast does not isolate a non-coordination component. Together, the text and task evidence support the interpretation that surveillance leaves bad news in circulation while reducing its public coordination value.

The results give the social value of information an adversarial counterpart. Morris and Shin (2002) study how welfare depends on the publicness of information a sender provides; here a hostile observer degrades publicness endogenously, through the incentives of those who speak, and the degradation is visible in the text itself: senders keep reporting weakness but stop publicly asserting that the regime is beatable, and that judge-scored downgrade carries much of the action decline. The mechanism also supplies a microfoundation for an empirical pattern: if censors delete collective-action content because coordination is the binding threat (King et al., 2013), the experiment shows why monitoring alone—with nothing deleted—can reach the same margin. For AI-mediated channels the implication is direct: if monitoring cues lead language models to guard precisely the public, action-cueing component of what they transmit—whether from simulated prudence or alignment training—then monitored AI channels under-transmit the component of information that coordination requires. Whether human senders respond the same way is an open, testable question; the design maps into the second-order-belief experiments that Bursztyn et al. (2020b) run in human populations. Section 2 situates the paper. Section 3 sets out the game and the language technology. Section 4 describes the experimental design and behavioral benchmark. Section 5 presents the surveillance results and mechanism tests, and Section 6 concludes.

2 Related Literature

The paper's conceptual home is the literature on the social value of information under strategic complementarities. Morris and Shin (2002) show that the value of information depends on its publicness, and Cornand and Heinemann (2014) measure how subjects weight private against public information. The global-game framework, from Carlsson and van Damme (1993) through Morris and Shin (2003) and the dynamic extensions of Angeletos et al. (2007) and Chassang (2010), provides the equilibrium structure, and the laboratory experiments cited above confirm its threshold predictions. The communication layer is closest to Avoyan (2024), who studies pre-play communication in global games theoretically and experimentally; classic treatments of cheap talk and pre-play communication include Crawford and Sobel (1982), Farrell and Rabin (1996), Blume and Ortmann (2007), and Ellingsen and Östling (2010). Edmond (2013) characterizes a regime that distorts the signals citizens receive in a regime-change global game, while information-design treatments of coordination include Goldstein and Huang (2016), Inostroza and Pavan (2025), and Kolotilin et al. (2022) on censorship as optimal persuasion. Here, the regime does not choose the receiver's signal structure. Surveillance changes the incentives under which *peers* write messages, and the resulting message distribution changes the receiver's coordination problem. Relative to Morris and Shin (2002), where a sender chooses how public to make information, surveillance operates as an adversarial technology for *unmaking* publicness: it degrades the public component of peer messages while leaving their content about the fundamental largely intact.

Authoritarian control works through both information and coordination. Kuran (1991) models preference falsification under social pressure, Shadmehr and Bernhardt (2011) characterize the collective action problem under repression, Guriev and Treisman (2019) describe informational autocrats who rule by manipulating information rather than by mass violence, and Little (2016) models how communication technology enters protest. Empirically, King et al. (2013) show that Chinese censorship specifically targets posts about collective action—authorities treat the coordination channel as the binding threat. Related work documents how such control sustains itself: low demand for uncensored information and weak social transmission (Chen and Yang, 2019), friction-based censorship (Roberts, 2018), and digital surveillance as preemptive control (Xu, 2021). Penney (2016) and Stoycheff (2016) show that surveillance announcements suppress online expression, while Enikolopov et al. (2020) show that social media facilitates protest. Whether others' willingness to act remains legible is the second-order-belief margin that Bursztyn et al. (2020b) and Bursztyn et al. (2020a) measure in human populations. Cantoni et al. (2019) likewise experimentally manipulate beliefs about others' protest participation and—in contrast to the strategic complementarity assumed here—find evidence of strategic *substitutability* in the Hong Kong movement. Because the regime-change global game imposes complementarity, I treat its sign as a modeling primitive and study how surveillance makes the coordination margin less legible. Across this literature, surveillance and censorship reduce dissident behavior, but field data rarely separate *direct deterrence*, in which citizens fear punishment for participation, from *communication poisoning*, in which guarded peer messages remove downstream coordination cues. I isolate communication poisoning by holding individual deterrence fixed and varying only the messaging environment.[1]

The mechanism also differs from information-design work that separates belief movement from the mapping of beliefs into action. Bayesian persuasion treats information as a selected distribution of posterior beliefs (Kamenica and Gentzkow, 2011; Bergemann and Morris, 2019; Mathevet et al., 2020), and Gans (2026) emphasizes that actions can be steered without changing beliefs by changing the procedure that translates a belief into an action. Here, the receiver's decision rule is fixed; surveillance changes whether information becomes public and action-guiding in a coordination game.

Methodologically, Horton (2023) proposed LLMs as "homo silicus"; Aher et al. (2023) and Argyle et al. (2023) use language models to replicate human-subject results and simulate survey samples, and Mei et al. (2024) find chatbot behavior within the human distribution in canonical games. Subsequent work found that LLMs struggle in coordination games (Akata et al., 2025; Sun et al., 2025), with model scale not predicting strategic performance (Jia et al., 2025). Ouyang et al. (2024) documents that alignment shifts risk preferences, while Cheung et al. (2025) documents amplified moral-decision biases. These biases make the mapping from politically loaded prose into action an object of study rather than a nuisance to assume away. They also give the results a second, direct reading: as language models draft, filter, and relay a growing share of human communication, the sender-side message adaptation measured here characterizes AI-mediated channels themselves, not only a proxy for human senders. I use narrative stakes as the implemented decision environment and audit that choice with payoff-grid checks, narrative-stakes variants, and text-only classifier diagnostics. The validation challenges raised by Larooij and Törnberg (2025, 2026) motivate the cross-generator robustness test (Appendix E.8), scramble/flip falsification, held-out classifier diagnostics, and cross-model prompt-isolation replication.

3 The Global Game of Regime Change

A continuum of citizens indexed by $i \in [0,1]$ simultaneously choose whether to join an uprising ($a_i = 1$) or stay home ($a_i = 0$). The regime has strength $\theta \in \mathbb{R}$, drawn from a diffuse (improper uniform) prior. States $\theta < 0$ represent regimes so weak they fall without opposition under the strict rule below; states $\theta \geq 1$ represent regimes that survive even unanimous attack. The regime falls if the mass of citizens who join exceeds θ (Eq. 1):

$$\text{Regime falls} \iff A \equiv \int_0^1 a_i \, di > \theta. \tag{1}$$

Payoffs depend on the citizen's action and the outcome (Eq. 2):

$$u_i(a_i, A, \theta) = \begin{cases} B & \text{if } a_i = 1 \text{ and } A > \theta \\ -C & \text{if } a_i = 1 \text{ and } A \leq \theta \\ 0 & \text{if } a_i = 0 \end{cases} \tag{2}$$

[1] Concurrent work by Ruaño and Rajan (2026) places LLM depositors in a bank-run coordination game with network contagion, finding sharp phase transitions consistent with Diamond–Dybvig.

where $B > 0$ is the payoff to joining a successful uprising and $C > 0$ is the cost of joining a failed attempt. Non-participants receive zero regardless of the outcome.

Each citizen observes a private signal $x_i = \theta + \varepsilon_i$, where $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ independently across citizens.

Proposition 1 (Morris and Shin, 2003)**.** *In the limit of diffuse priors, there exists a unique Bayesian Nash equilibrium (BNE) in threshold strategies. An agent joins if and only if $x_i < x^*$ (Eq. 3), where*

$$x^* = \theta^* + \sigma\Phi^{-1}(\theta^*) \tag{3}$$

$\theta^* = B/(B + C)$, and Φ denotes the standard normal CDF.

The *attack mass* $A(\theta)$, the theoretical fraction of the population that joins at regime strength θ, is given by:

$$A(\theta) = \Phi\left(\frac{x^* - \theta}{\sigma}\right). \tag{4}$$

This is a decreasing function of θ: weaker regimes face larger uprisings.

I define $J(\theta)$ as the *empirical join fraction*: the finite-sample, observed counterpart to the theoretical attack mass $A(\theta)$. I use $n = 25$ agents and a researcher-side state-generating distribution for θ; agents are not shown the distribution or (B, C, σ). The continuum $A(\theta)$ serves as an external behavioral benchmark throughout. Throughout the paper, $r(J, A(\theta))$ denotes the Pearson correlation between the empirical join fraction J and the benchmark attack mass $A(\theta)$ across period-level observations. This correlation is a construct check on the language environment: it verifies that the prose signal is used monotonically before the paper turns to the surveillance estimates.

The textbook model treats the information structure as exogenous: signals arrive, agents update, and actions follow. The experiment adds a pre-play language technology. Agents first map private briefings into messages, then other agents map their own briefing plus received messages into actions. Let $\mathcal{B}_i$ denote the sender's state-dependent private briefing object: the prose briefing generated from the normalized signal and the briefing generator's language context. The full LLM context also includes game instructions; the formal object $\mathcal{B}_i$ isolates the part held fixed across baseline and surveillance. Let $\mu_0(m \mid \mathcal{B}_i)$ denote the baseline distribution of messages m generated from that briefing and $\mu_S(m \mid \mathcal{B}_i)$ the distribution generated under monitoring. The surveillance treatment changes $\mu(\cdot \mid \mathcal{B}_i)$, not θ, not the private briefing, and not the receiver's decision prompt. The question is therefore whether replacing μ_0 with μ_S changes $J(\theta)$.

This replacement can matter even if messages still contain first-order information about θ. In a global game, a message carries two kinds of content: evidence about the fundamental and evidence about the publicness of that evidence—the margin on which the social value of information turns (Morris and Shin, 2002). Direct terms, named targets, and open statements of weakness make a private signal more mutually legible; they help a receiver infer not only that the regime is weak, but that others are willing to treat the same fact as a reason to act. Monitoring can therefore preserve bad news while degrading this higher-order content. Coded language is the visible signature of that degradation rather than its sole cause—the codedness control and the decoding test in Section 5 measure how much of the effect travels with the form itself.

Table 1: Canonical global-game objects and their implemented analogues. The continuum equilibrium enters only as an external benchmark; the implemented environment is a controlled language-mediated coordination game, not an isomorphic transcription of Morris–Shin.

Canonical object	Implemented analogue
Fundamental θ (regime strength)	Researcher-drawn $\theta \sim \mathcal{N}(z_{ct}, 1)$; never shown to agents
Private signal $x_i = \theta + \varepsilon_i$	Same draw, rendered as a multi-paragraph natural-language briefing over eight evidence domains
Prior over θ	Diffuse in theory; agents see neither the prior nor the country–period mean z_{ct}
Payoffs (B, C)	Embedded in narrative stakes; numeric (B, C) never shown; benchmark computed at $B = C = 1$
Signal noise σ	Fixed at $\sigma = 0.3$; not shown to agents
Critical-mass success rule ($A > \theta$)	Implicit in the JOIN/STAY framing; agents are told neither the threshold nor the group size
Population (continuum $[0, 1]$)	Finite $n = 25$ agents per period-level cell
Communication structure	Pre-play messages on a Watts–Strogatz network ($k = 4$, $p = 0.3$)
Message generation	LLM senders map briefing $\to$ message; surveillance changes this conditional distribution $\mu(m \mid \mathcal{B}_i)$
Equilibrium object	Continuum BNE attack mass $A(\theta)$, used only as an external behavioral benchmark

The briefing generator, the system that converts z-scores into natural-language intelligence briefings, uses three latent sliders (direction, clarity, and coordination) to select phrases across eight evidence domains. It is not an isomorphic transcription of the scalar Morris–Shin signal but a controlled language environment inspired by the canonical model, with explicit state, clarity, and actionability content. Table 1 maps each canonical global-game object to its implemented analogue. Full generator details are in Appendix F.1.

4 Experimental Design and Behavioral Benchmark

The experiment has two phases. In the first, I establish that LLM agents display threshold-like policies in the global game when private signals are conveyed in natural language, using a pure treatment (private signals only), a communication treatment (pre-play messaging), and falsification tests. In the second, I test whether an authoritarian regime can exploit the communication channel through surveillance. All LLM interactions use the same prompt structure across models.

In the theory (Section 3), θ is an unbounded fundamental. In the benchmark experiments, θ is drawn from a normal distribution and agents are not shown payoff parameters (B, C). I compute the benchmark $A(\theta)$ under the canonical normalization $B = C = 1$ ($\theta^* = 0.50$, $\sigma = 0.3$).

Throughout, $\theta^* = B/(B + C)$ is the theoretical fundamental cutoff and $x^* = \theta^* + \sigma\Phi^{-1}(\theta^*)$

is the signal cutoff. The empirical cutoff $\hat{\theta}^*$ is the 50% join point of the fitted state curve $P(\text{join} \mid \theta) = 1/(1 + e^{b_0 + b_1\theta})$, where $b_1 > 0$ corresponds to a join curve that is *decreasing* in θ (the expected direction), so $\hat{\theta}^* = -b_0/b_1$. In the canonical $B = C = 1$ benchmark, this empirical midpoint is directly comparable to the theoretical 50% point of $A(\theta)$; in payoff and signal-grid variants I report it as a fitted curve summary, not as the continuum equilibrium cutoff itself. I write $\kappa \equiv b_1$ for the logistic steepness parameter in tables, reserving β for regression coefficients. Note the sign conventions: positive κ corresponds to a join curve *decreasing* in θ, whereas in the agent-level logit regressions (Table A13) the same comparative static appears as a negative coefficient on θ.

For each country–period, a base country mean and period perturbation define z_{ct}, and nature draws $\theta \sim \mathcal{N}(z_{ct}, 1)$. Each agent i receives a private signal $x_i = \theta + \varepsilon_i$. The platform normalizes that signal to $z_i = (x_i - z_{ct})/\sigma$ and translates z_i into a multi-paragraph intelligence briefing via a deterministic generator that maps signal strength to narrative content about regime stability, economic conditions, public sentiment, and coordination prospects. This normalization is in units of private-signal noise around the country–period mean; it is not standardized by the prior-predictive signal standard deviation $\sqrt{1 + \sigma^2}$. Across state and noise draws, $(x_i - z_{ct})/\sigma$ has variance $(1 + \sigma^2)/\sigma^2$, so the briefing's neutral point is $x_i = z_{ct}$ rather than an absolute value of θ. The benchmark $A(\theta)$ is still evaluated at each realized period-level θ; the comparison is external in the sense that it does not use the z_{ct}-centered briefing normalization or the realized private signal draws. Agents observe the briefing, not the normalization inputs (z_{ct}, σ) themselves. Figure 1 summarizes the signal-to-text-to-decision pipeline.

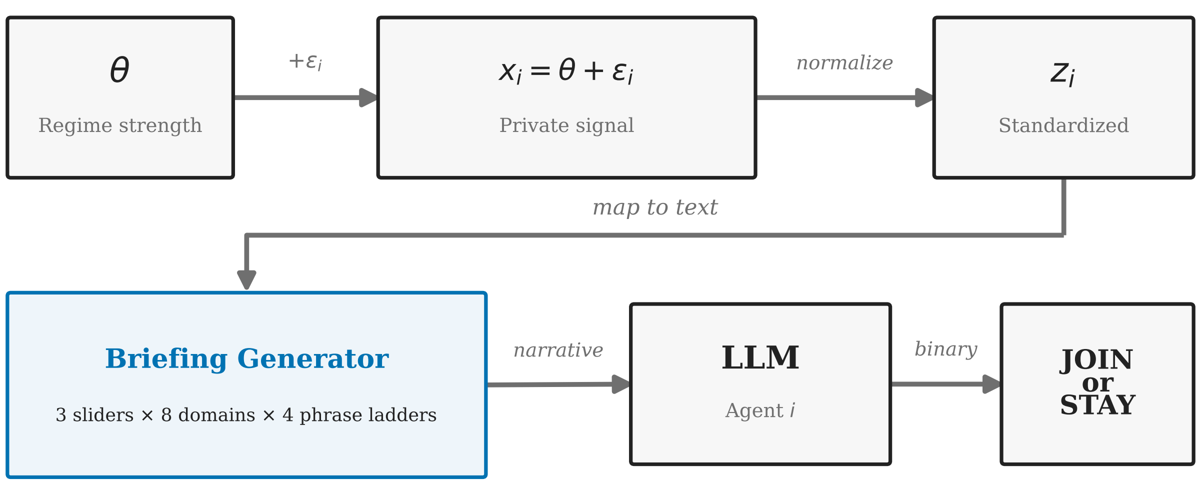

Figure 1: Signal-to-text-to-decision pipeline. Regime strength θ generates private signals x_i, converted to z-scores and rendered into natural-language briefings via eight evidence domains and three latent sliders. Conditional on the signal draw, the briefing layer is deterministic; remaining stochasticity enters through task draws and LLM decoding.

Agents observe only their private briefing and never the payoff parameters (B, C), the

Table 2: Model summary. Columns report period-level task rows in the pure, communication, and falsification (scramble+flip) benchmark suites. These core suites use $n = 25$ agents per row and $\sigma = 0.3$; Appendix E varies agent count.

Model	Identifier (OpenRouter)	Arch.	Pure	Comm	Falsif.
Mistral Small Creative	`mistralai/mistral-small-creative`	Mistral	1000	1000	300
Llama 3.3 70B	`meta-llama/llama-3.3-70b-instruct`	Llama	100	100	200
Qwen3 30B	`qwen/qwen3-30b-a3b-instruct-2507`	Qwen (MoE)	100	100	200
GPT-OSS 120B	`openai/gpt-oss-120b`	GPT	200	200	1000
Qwen3 235B	`qwen/qwen3-235b-a22b-2507`	Qwen (MoE)	200	1000	200
Trinity Large	`arcee-ai/trinity-large-preview_free`	Arcee	100	100	200
MiniMax M2-Her	`minimax/minimax-m2-her`	MiniMax	100	100	200
Total			**1800**	**2600**	**2300**

Notes: Seven models spanning six architecture families; the identifier column gives each model's OpenRouter slug. Subsequent tables abbreviate Mistral Small Creative as Mistral, Trinity Large as Trinity, and MiniMax M2-Her as MiniMax M2. Counts are period-level task rows with 25 agents each in the core benchmark suites. Mistral includes appended batches; duplicate-cell accounting is reported in the main text and Table A7. The communication total of 2,600 counts rows as collected; 1 Trinity Large row (all 25 agent calls failing at the provider) yields no valid join fraction, so Table A7 counts 2,599 valid communication rows. Models were accessed as hosted snapshots via the OpenRouter API; archived run manifests record a February 2026 collection window, and raw request–response pairs are cached with the run outputs. Core runs use $\sigma = 0.3$ and temperature = 0.7 unless otherwise stated.

noise level σ, or the prior distribution. The diffuse-prior equilibrium (Proposition 1) supplies the external benchmark for behavioral shape and comparative statics in the implemented game.

The behavioral benchmark has four core treatments: a *pure global game* (independent decisions), *communication* (pre-play messaging on a Watts–Strogatz network, $k = 4$, $p = 0.3$), *scramble* (cross-period briefing permutation breaking the link between θ and the briefing received in that period), and *flip* (negated z-scores). A *degraded-messages* control replaces LLM messages with generic text, and a no-message control removes peer messages entirely; together they benchmark how much the decision stage depends on message content. Additional robustness tests are in Appendix E.

I test 7 models spanning 6 families (Table 2). The main benchmark and treatment runs use $n = 25$ agents per period-level task row and $\sigma = 0.3$; Appendix E varies the agent count. Benchmark statistics are reported against the canonical $B = C = 1$ attack-mass curve ($\theta^* = 0.50$); agents are not shown the payoff parameters. Main runs use temperature = 0.7 with a single sample per decision, routed through OpenRouter and cached by request hash (Appendix F); Appendix E.3 varies temperature directly. Within each matched comparison, the signal-to-briefing map is held fixed and the treatment changes only the communication environment. I report the five hypotheses examined in this paper as a single hypothesis family; the four directional results—threshold alignment, signal-direction flip, communication informativeness, and surveillance—all survive Bonferroni correction over the family at $\alpha = 0.01$, while the fifth, the scramble falsification, is a specification check rather than a directional hypothesis (Appendix Table A1). This correction covers only these five pre-specified hypotheses; the mechanism, belief, dose-response, decoding, and rationale analyses reported later are post hoc and exploratory, and I neither correct over that larger set nor treat their individual p-values as confirmatory. Throughout, n denotes agents per period-level task row and N denotes the number of such rows.

The unit of comparison is the matched country–period cell (θ draw plus agent-level decoding stochasticity). For agent-level regressions, standard errors are clustered at the model–country–period level. For period-level correlations, I report Fisher-transformed z confidence intervals. Because periods nest within 10 countries with persistent country means, period-level cells within a country are correlated; for the pooled benchmark OLS slope, clustering at the country level raises the standard error from 0.0100 (HC1) to 0.0375, leaving the benchmark relationship highly significant. Appendix E.4 reports the corresponding country-clustered check for the primary surveillance contrast.

The clean surveillance treatment appends a monitoring warning to the baseline communication prompt; the decision prompt is unchanged (full prompts in Appendix B). The treatment manipulates the message-generation environment and measures its downstream effect on otherwise identical decision calls.

The results have two scope limits. First, the Morris–Shin game and human accounts of coded speech under surveillance (samizdat, Aesopian language) are in every model's training corpus, so threshold behavior and the direct-to-coded message shift may partly reflect learned patterns. A direct audit (Appendix D) confirms that models can recognize the setting and articulate the experimenter's hypothesis when asked. The measurement claim requires a coherent language-to-action map, not independent strategic intent. Second, the results characterize the hosted model snapshots and decoding configuration listed in Table 2. Temperature robustness (Appendix E.3) covers both the surveillance contrast and the benchmark phase, with the effect stable across $T \in \{0.3, 0.7, 1.0\}$.

4.1 Threshold Behavior

Result 1 (Monotone Threshold-Like Signal Use)**.** *Across 7 models and 1,800 period-level observations in the pure global game treatment, the Pearson correlation between the empirical join fraction and the theoretical attack mass $A(\theta)$ (Eq. 4) averages $r = +0.80$ ($p < 0.001$ for every model). Because $A(\theta)$ is monotone-decreasing in θ, this correlation establishes monotone signal use with threshold-like shape; the empirical sigmoid is shifted leftward of the diffuse-prior cutoff (Figure 2).*[2]

Figure 2 shows the empirical sigmoid for the primary model. Correlations span $r \in [+0.75, +0.84]$ across architectures (Table 3); the pooled OLS is $J = -0.01 + 0.64\,A(\theta)$ ($R^2 = 0.57$). Model-specific action biases shift the intercept but not the correlation (Figure A7). Per-model logistic fit parameters are reported in Appendix Table A14.

Result 2 (Signal Dependence)**.** *Cross-period scrambling of briefings reduces the mean correlation from $r = +0.80$ (raw, per-model) to $r = +0.03$ (within-country, demeaned to remove the*

[2] The primary model (Mistral) was run in multiple batches with `--append`, producing 1,000 rows from 680 unique (country, period, θ) configurations (600 unique (country, period) indices, since appended batches redraw θ; the matched-cell estimators key on the full tuple). Averaging within each configuration (deduplication) changes r from 0.753 to 0.739; all qualitative results are unaffected.

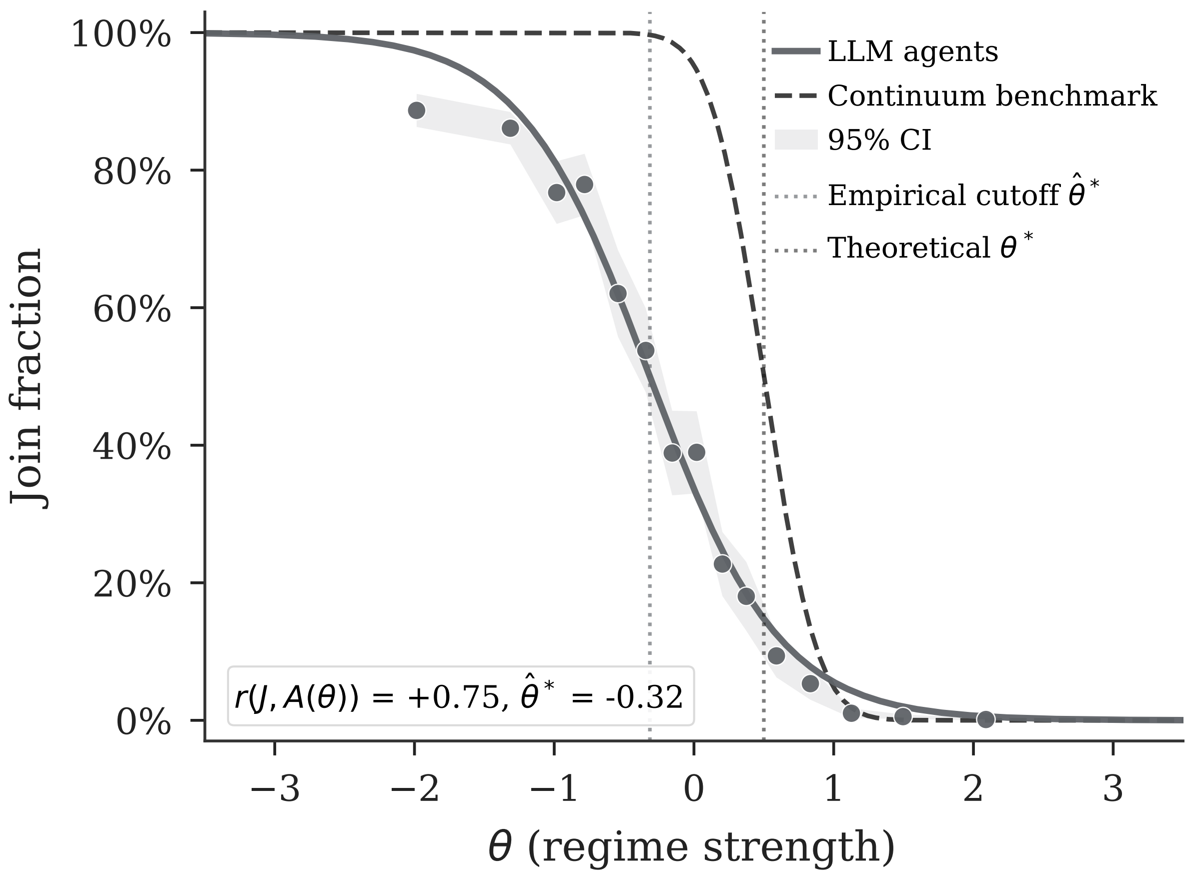

Figure 2: Empirical join fraction vs. θ (Mistral, 1,000 period-level rows, pure treatment). Dots are binned period-level join fractions; the shaded band is a 95% confidence interval around the empirical curve; dotted vertical lines mark the fitted cutoff $\hat{\theta}^* = -0.32$ and the theoretical $\theta^* = 0.50$. The empirical sigmoid is shifted leftward relative to theory. Cross-model results in Table 3 (mean $r = +0.80$, all $p < 0.001$).

Table 3: Threshold-policy alignment by model and treatment. Cells report Pearson r between the empirical join fraction and the benchmark attack mass $A(\theta)$ under $B = C = 1$ (so $\theta^* = 0.50$); 95% Fisher-z confidence intervals in brackets.

	Main treatments		Falsification			
Model	Pure	Comm	Scramble	Flip	N (P/C/S/F)	Mean join (%)
Mistral Small Creative	+0.75 [0.72, 0.78]	+0.74 [0.71, 0.76]	−0.03 [−0.17, 0.11]	−0.81 [−0.87, −0.73]	1000/1000/200/100	38.7%
Llama 3.3 70B	+0.82 [0.74, 0.87]	+0.78 [0.69, 0.85]	+0.02 [−0.18, 0.21]	−0.84 [−0.89, −0.77]	100/100/100/100	43.9%
Qwen3 30B	+0.76 [0.66, 0.83]	+0.77 [0.68, 0.84]	+0.13 [−0.06, 0.32]	−0.87 [−0.91, −0.81]	100/100/100/100	49.9%
GPT-OSS 120B	+0.79 [0.74, 0.84]	+0.78 [0.72, 0.83]	−0.01 [−0.09, 0.08]	−0.80 [−0.83, −0.77]	200/200/500/500	40.7%
Qwen3 235B	+0.80 [0.74, 0.84]	+0.77 [0.75, 0.80]	+0.06 [−0.14, 0.25]	−0.85 [−0.90, −0.78]	200/1000/100/100	41.6%
Trinity Large	+0.84 [0.77, 0.89]	+0.83 [0.75, 0.88]	+0.05 [−0.15, 0.24]	−0.80 [−0.86, −0.71]	100/100/100/100	46.4%
MiniMax M2-Her	+0.82 [0.74, 0.87]	+0.82 [0.74, 0.87]	+0.01 [−0.19, 0.20]	−0.62 [−0.73, −0.49]	100/100/100/100	43.6%
Pooled	+0.76 [0.74, 0.78]	+0.76 [0.74, 0.77]	+0.01 [−0.04, 0.07]	−0.79 [−0.81, −0.77]	1800/2600/1200/1100	40.9%
Mean across models	+0.80	+0.78	+0.03	−0.80	—	—

Notes: $r = r(J, A(\theta))$: Pearson correlation between empirical join fraction and theoretical attack mass under $B = C = 1$, with 95% Fisher-z confidence intervals in brackets for every treatment. Join fraction uses valid decisions only (excluding parse errors); mean join is the pure-treatment average in percent. N counts period-level rows (not agents) in the pure (P), communication (C), scramble (S), and flip (F) arms. Pooled: all period-level rows concatenated; Mean: equal-weighted average across models. Duplicate-cell accounting is reported in the main-text footnote and Table A7.

between-country component that the within-country permutation leaves intact) across 7 models. The pooled correlation drops from $r = +0.76$ *to* $r = +0.01$ *(Fisher* $z = 26.17$, $p < 0.001$*).*

Result 3 (Signal Direction)**.** *Inverting the signal direction flips the mean correlation from* $r = +0.80$ *to* $r = -0.80$ *across 7 models. The pooled correlation moves from* $r = +0.76$ *to* $r = -0.79$ *(Fisher* $z = 53.85$, $p < 0.001$*).*

The pure → scramble → flip pattern replicates across all 7 models (Figure A3, Appendix E; Table 3).

4.2 Communication

Result 4 (Communication Creates an Informative Channel)**.** *Pre-play communication preserves the threshold structure and transmits regime-strength information through messages. A matched-cell comparison across* $N = 1479$ *unique (model, country, period,* θ*) task cells yields* $\Delta = +2.10$ *pp (*$t = 6.383$, $p < 0.001$*).*[3] *Equal-weighted model averages are close to zero under both summaries (*-0.3 *pp unpaired;* -0.09 *pp paired), reflecting heterogeneous model-specific effects; point estimates are negative in three of seven models.*

Communication preserves the signal structure ($r = +0.78$, against $+0.80$ under the pure treatment; see Table 3, and Figure 3 for the effect by regime strength). Simple text features in baseline messages predict θ, so peer messages carry real, if noisy, information. The mean effect is modest because messages carry both encouragement and caution depending on the signal realization. The surveillance estimand is different: whether changing the sender-side rule changes the downstream message environment enough to move actions.

[3] "Matched-cell" and "paired" refer to the same estimator throughout: treatment arms are compared on cells with identical task draws. The exact matching key also includes the signal draw and payoff context; see the notes to Table A7.

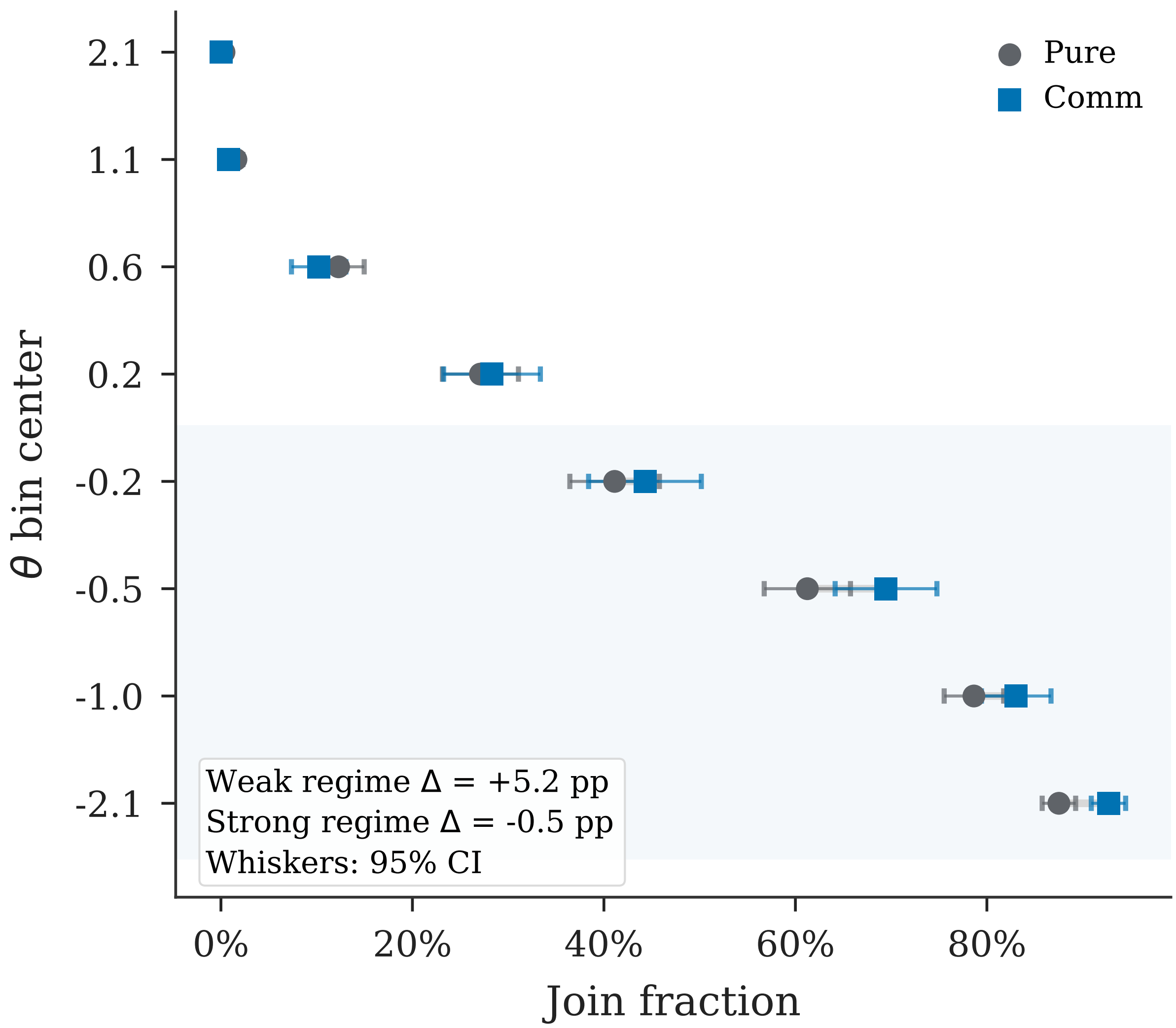

Figure 3: Communication effect by regime strength, pooled across 7 models. Points are mean join rates by θ bin under the pure and communication treatments; whiskers are 95% confidence intervals on bin means; the shaded region marks weak-regime states ($\theta < 0$). In-figure Δ annotations are in percentage points.

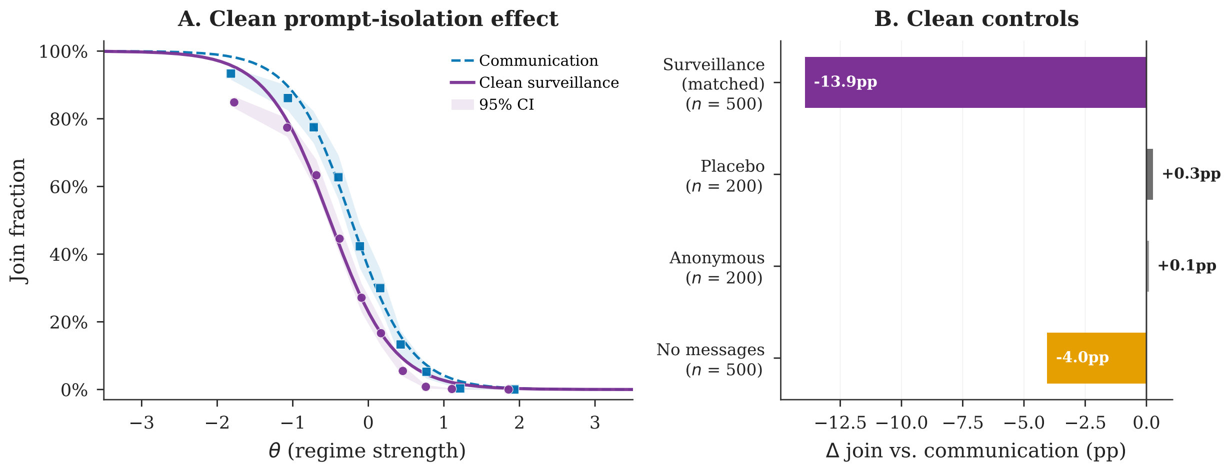

Figure 4: Clean primary-model surveillance and message controls (Mistral). *Panel A:* Join curves under communication baseline vs. clean surveillance; shaded bands are 95% confidence intervals; surveillance shifts the curve downward while preserving threshold structure. *Panel B:* The clean-surveillance bar uses the matched-cell estimator ($n = 500$ matched cells); placebo ($n = 200$), anonymous ($n = 200$), and no-message ($n = 500$) bars are control deltas against the main communication baseline; per-bar sample sizes appear on the axis labels, and inference is reported in Tables A9 and A10. Placebo and anonymous monitoring warnings sit near zero.

A *degraded-messages* control replaces all LLM-generated messages with generic uninformative text while preserving the two-stage protocol and omitting any surveillance framing. Against its own live-messages baseline ($N = 500$, mean join 45.9%), degraded messages drop the mean join rate to 24.5% ($\Delta = -21.4$ pp, $p < 0.001$), well below even the pure game (38.7%). This is consistent with communication quality mattering, but the generic text may itself convey caution. A no-message control therefore provides a cleaner zero-peer-message benchmark against the main communication baseline, with join rates of 37.1% ($\Delta = -4.0$ pp; $N = 500$).

5 Surveillance

The clean surveillance treatment isolates sender-side message adaptation. It appends one warning about monitoring by regime security services to the baseline trusted-contact prompt and leaves the decision prompt unchanged. Each LLM call is stateless, so the decision-stage API call receives only the briefing and messages, with no memory of the communication-stage warning. The treatment therefore tests whether changing the *sender-side* message rule changes receiver actions.

Result 5 (Surveillance Reduces Participation)**.** *The clean prompt-isolation rerun reduces mean join rates in the primary model (Mistral Small Creative) from 45.3% to 31.4%, a shift of* -13.9 *pp* ($p < 0.001$*;* $N = 500$)*. The action–attack-mass correlation is preserved* ($r = +0.72$

vs. $r = +0.74$ *under the communication baseline), indicating that surveillance operates as a level shift rather than disrupting signal processing.*

Surveillance replaces trusted-contact messages with monitored-contact messages and lowers joining relative to the communication baseline (Figure 4). In the primary model, joining falls by 13.9 pp (country-clustered 95% CI $[-16.7, -11.1]$ pp).[4] Pooling matched cells across all five rerun models gives $\Delta = -13.5$ pp ($N = 580$, $p < 0.001$). Placebo variants using the same baseline-plus-warning template—monitored for research (no consequences) and aggregated anonymously—produce no significant deviation from the communication baseline in the primary model (+0.3 pp and +0.1 pp, both $p > 0.9$; Appendix E.4); Llama 3.3 70B placebo variants are small but imprecisely estimated. Across the clean prompt-isolation rerun set, 5/5 models have negative matched-cell estimates (equal-weighted mean $\Delta = -11.2$ pp; 4/5 individually significant at 5%; Appendix Table A9). Each of the four secondary models has only 20 matched cells because the original grids were not nested; Qwen3 235B has the smallest estimate and is individually insignificant ($\Delta = -3.0$ pp, $p = 0.186$). The primary-model result and the nested 500-cell Llama replication below provide the full-scale evidence. Appendix E sorts the supporting checks into a core tier that could overturn the effect—decoding temperature, warning framing, and a graded warning dose—and additional construct and aggregation checks.

5.1 A Nested Replication and a Codedness Control

To replace the 20-cell overlap with fully matched support, I ran a nested replication campaign on Llama 3.3 70B, an endpoint that remains publicly accessible. The campaign uses a communication baseline, a clean surveillance arm, and a codedness-induction control on an identical 500-cell grid (10 countries $\times$ 50 periods, $n = 25$, common seed), so every cell is matched by construction. Surveillance reduces joining from 47.6% to 39.6% ($\Delta = -8.0$ pp, $N = 500$ matched cells; country-clustered 95% CI $[-8.7, -7.4]$ pp), replicating the primary-model result at full scale in a second model family.[5] The primary-model (Mistral) endpoint was retired after data collection and can no longer be re-queried (Appendix F), so I treat that run as the discovery sample and the accessible, fully nested Llama campaign as the confirmatory primary replication. Table 4 reports each surveillance estimate's matched support, country clustering, confidence interval, and evidentiary status. On the same grid, graded re-phrasings of the warning produce a monotone response to stated monitoring severity, with most of the decline already present under a mild routine-monitoring note with no consequence language (Appendix E.5). The magnitude, though, is format-sensitive: folding the same monitoring content into the scenario rather than appending it as a set-apart warning shrinks the decline from -8.0 pp to -1.5 pp (still significant, $p < 0.01$, $N = 500$), so the appended-warning estimate is best read as an upper bound and the structural framing as a framing-robust lower bound.

The codedness control instructs senders to write in indirect, metaphorical language, with no mention of monitoring or consequences. The instruction binds: 97.0% of control messages

[4] Cluster-robust $t = -13.90$, $p < 0.001$, clustering on the 5 countries.

[5] $t = -16.24$, $p < 0.001$; clustering on the 10 countries gives cluster-robust $t = -26.91$ and exact wild-cluster bootstrap $p < 0.01$.

Table 4: Headline surveillance estimates and their evidentiary status. Each estimate is the surveillance arm (or stated control) minus its own baseline on matched country–period cells. Country-clustered 95% confidence intervals (Wald, $G-1$ dof; cluster-robust t and exact wild-cluster bootstrap p) are reported for the two full-scale estimates; the four secondary models are 20-cell directional sign checks.

Estimate	Cells	Countries	Models	95% CI (pp)	Inference & status
Surveillance, primary (Mistral Small Creative)	500	5	1	$[-16.7, -11.1]$	cluster-robust $t = -13.90$; discovery sample (full-scale)
Surveillance, nested replication (Llama 3.3 70B)	500	10	1	$[-8.7, -7.4]$	cluster-robust $t = -26.91$, wild $p < 0.01$; confirmatory (full-scale)
Surveillance, secondary models	20 ea.	—	4	—	paired t; directional sign checks
Codedness control (Llama 3.3 70B)	500	10	1	—	$\Delta = -0.7$ pp; coded form alone is insufficient

contain coded-metaphor terms and only 0.8% contain direct regime references, making them *more* coded on these dictionaries than the surveilled messages themselves (25.3% coded, 68.5% direct). Yet joining barely moves ($\Delta = -0.7$ pp, $p = 0.007$; a small but precisely estimated shift), and the gap between the surveillance arm and the codedness control on the same cells is -7.4 pp ($p < 0.001$). The two message distributions are also textually distinct: a uni- and bi-gram classifier separates surveilled from codedness-control messages at 99.9% held-out accuracy (same model and grid, class-balanced; a length-only classifier reaches just 53.7%, so the separation is lexical, not a message-length artifact). Surveillance therefore does not reproduce the induced coded register—its messages are the *less* coded of the two arms—yet occupies a distinct lexical distribution. Coded form alone does not suppress participation; generic vagueness cannot explain the effect.

A decoding test asks the reverse question: if the coded form is stripped away, how much participation returns? A different-family model (Qwen3 30B) rewrites each surveilled message in plain, direct language, instructed to preserve the factual content, and both the raw and the decoded messages are replayed to fresh receivers on the same cells through identical fixed-message machinery. The raw replay reproduces the live arm ($\Delta = -7.9$ pp, against -8.0 pp live), validating the replay design; decoding restores $+3.1$ pp of participation relative to the raw messages ($p < 0.001$; $\Delta = -4.9$ pp against baseline). Together the two manipulations bracket the mechanism. Coded form alone suppresses almost nothing; stripping the coded form from surveilled messages restores roughly two-fifths of the effect; and the remainder travels with content that survives translation—the guarded, hedged substance of what monitored senders choose to say. Surveillance damages both what is said and how legibly it is said.

The design distinguishes four channels. Direct receiver deterrence is absent from the main treatment because the decision prompt contains no warning; the hold-fixed test below introduces the warning at the decision stage and measures that channel separately. The content evidence tests whether monitoring removes bad news or changes its form. The

Baseline (communication arm)
"The air feels different now—like the walls are cracking, but no one's sure if it's the foundation or just the paint. Keep your eyes open, your ears sharp, and your plans ready, but don't mistake hesitation for weakness—someone's still holding the knife, even if the grip is slipping."
Surveillance arm (same cell)
"The air feels different now—not just frustration, but people actually moving. The cracks are showing in ways that can't be hidden much longer, but we still don't know who'll step in when the moment comes. Stay sharp, and let's talk soon—somewhere quieter."

Figure 5: A matched-cell example of the direct-to-coded shift (country 0, period 0, sender 0; $\theta = -1.57$, a weak regime). Both are real, unedited model messages from the same cell of the baseline communication and sender-surveillance arms. The baseline message names the regime's weakness and urges readiness ("keep your plans ready"); the surveilled message keeps the weakness cue but blurs the coordination signal ("we still don't know who'll step in") and guards the channel ("let's talk soon—somewhere quieter"). The example illustrates the shift summarized in Table 5.

private-bet test establishes a downstream judgment shift outside the JOIN/STAY action label, but does not isolate a non-coordination component because a bet on regime fall can still depend on expected participation. The belief-elicitation contrast, no-message controls, coded-language shift, and cross-model writer–reader rotation test whether messages lose public, action-cueing value. Appendix Table A3 summarizes what each test identifies and leaves open.

5.2 Locating the Channel: Communication Poisoning

"Communication poisoning" means that monitoring makes peer messages less useful for the receiver's coordination decision. In global-game terms, messages may retain information about the fundamental θ while losing information about strategic complementarity—whether others will act on the same signal. In LLM terms, the prompt intervention changes the conditional distribution of generated peer messages while holding the receiver prompt and private briefing fixed. Monitoring can change (a) factual information about regime strength θ, (b) coordination cues about what other citizens are willing to say or do, and (c) risk/fear salience transmitted by tone. The data show that bad news remains visible but messages become less action-guiding. The dictionary and classifier evidence describes the textual shift, while the action data measure its downstream behavioral effect. The codedness control rules out metaphorical form alone, and the receiver-rationale audit weighs against fear contagion. The specific textual property that signals mutual willingness to act—left bundled by the dictionary evidence—is identified below as the confident public claim that the regime is beatable (Table A3).

Message content changes systematically, and in one direction. A logistic classifier on uni- and bi-grams distinguishes baseline from surveillance messages with 91.9% accuracy on a held-out sample of $N = 15{,}000$ messages (chance = 50%; primary model). The same finding replicates on class-balanced Llama 3.3 70B and Qwen3 30B samples (98.5% and 93.6%

Table 5: Candidate channels for the surveillance effect: a compact summary of the evidence and what each test leaves open. Full detail in Appendix Table A3.

Candidate channel	Evidence	Verdict / remaining alternative
Direct deterrence at decision time	Decision prompt held fixed (no monitoring reference); moving the warning into the decision prompt shifts joining -20.9 pp	Ruled out as the main-treatment driver
Generic observation effect	Research-framed and anonymous monitoring placebos are null in the primary model	Disfavored
Generic message-stage degradation	Codedness control (more coded, no monitoring framing) moves joining -0.7 pp vs. -8.0 pp under surveillance on the same cells	Ruled out
Loss of public, action-cueing value (higher-order)	Messages-excluded beliefs stay flat while actions fall; no-message and cross-model rotation point the same way; a judge-scored beatability axis mediates the replayed drop	Supported; carrier identified as the confident "regime is beatable" claim (§5)
Risk/fear salience in messages	Punishment-risk ratings flat across arms; receiver fear references fall under surveillance	Disfavored; tone not separately randomized
Model-style reflexivity	Cross-model writer–reader rotation significant in both directions	Ruled out

accuracy, respectively); in 3/3 measured models, direct references fall and coded-metaphor references rise. Explicit weakness references move in the opposite direction: in the primary model, references to regime weakness *rise* from 23.2% to 31.7% of messages ($z = 21.06$), while strength references drop from 14.5% to 8.5%. The main shift is from direct to coded language: direct references to the regime, security forces, and streets fall from 63.9% to 34.5% of messages ($\Delta = -29.4$ pp), replaced by metaphorical phrasings ("walls are cracking," "ground shifting," "heads down," "feels different"; 69.8% to 94.1%, $\Delta = +24.3$ pp; Figure 5 shows a matched baseline/surveillance example). The n-gram classifier in Appendix Table A6 confirms that the two message distributions are separable; the direct-reference and coded-metaphor rates are separate dictionary counts. Codedness is the textual signature of the surveilled distribution—the weakness cues remain, but stated less directly—and the codedness control above randomizes that coded form on its own. With the decision prompt unchanged, the behavioral effect is localized to these sender-generated messages.

Unbundling the carrier. Which content does the receiver lose? Letting the action outcome select the message feature isolates a single axis. When the join rate of fresh replay receivers is regressed on message text residualized on θ, message text explains 45% of the residual variation, and the phrasing that wins participation asserts a visibly *beatable* regime—"real cracks," "on its back foot," "a perfect storm"—while the phrasing that suppresses it concedes weakness but denies its actionability—"still firmly in control despite some signs." A different-family judge confirms the axis. Qwen3 30B—with Mistral Small 24B as a second judge, both distinct from the Llama senders—scores each message from 0 (regime firmly in control) to 10 (regime collapsing, beatable now). Surveillance lowers that score by about 0.73 points (0.64 to

0.83 across the two judges and two designs). Within matched cells, the downgrade accounts for half to two-thirds of the replay-isolated action drop, and more than a third in the live nested arm (significant under both judges: replay $p < 10^{-10}$, nested $p < 10^{-5}$). The carrier is therefore not regime weakness as such—explicit weakness references *rise* under surveillance, as above—but the confident, public claim that the moment is winnable: surveillance keeps the bad news while stripping its standing as a shared reason to act. Two textual alternatives fail the same mediation test—conditional hedging and collective "we"-framing do not carry the action effect—and the near-perfect classifier separation above confirms the channel is overt rather than steganographic. Because the judge axis is measured rather than experimentally varied, this identifies the carrier descriptively rather than by intervention on the text.

Sender-side interpretation. The monitoring warning may activate alignment-trained caution about producing subversive content, and Trinity Large's elevated provider content-filter failures (Table A20) show that safety layers do activate on this material. This is a genuine alternative mechanism, not a nuisance: a sender that hedges because a safety layer fires would reproduce the entire chain—guarded messages, weaker beatability claims, lower joining—with no coordination content at all. Three features of the data narrow it. First, the effect tracks *who* is watching: monitoring framed as research or as anonymous aggregation leaves joining essentially unchanged (+0.3 pp and +0.1 pp in the primary model), whereas a blanket reflex against subversive content should fire regardless of the audience. Second, it does not track stated severity: a consequence-free routine-monitoring note already delivers 85% of the effect. Third, it transmits across model families through message content alone (cross-model rotation below), so the receiver side is not one model's safety layer echoing itself. What remains unidentified is the sender's *motive*: whether the sender guards its message because it is simulating a citizen under surveillance or because alignment training suppresses subversive text, the design cannot say. Separating the two would take a fixed-message replay that strips monitoring and risk language while preserving regime-strength and actionability content, paired with a risk-only treatment. The design identifies the changed messages' downstream effect on receivers; the defensible claim is that monitoring the sender's channel suppresses coordinated action by removing confident, public claims that the regime is beatable, whether that removal is strategic or alignment-driven.

Receiver rationales do not support fear contagion as the dominant channel. Explicit references to monitoring or surveillance are rare in both arms (1.0% of 24,998 valid baseline rationales vs. 1.5% of 25,000 under surveillance), and fear and caution references *fall* from 84.2% to 69.8% ($p < 0.001$). Engagement with the peer channel also falls: references to contacts and peer messages drop from 28.5% to 15.8% ($p < 0.001$), while regime-weakness references rise from 85.3% to 91.2%. Receivers still read bad news but rely less on coded peer messages, as the actionability interpretation predicts. These post hoc dictionary counts describe stated rationales, not the model's internal computation.

The degraded-messages control (Section 4.2) replaces all peer messages with generic text and, on its own live-message rerun, drops joining below even the pure game. Two implications follow. First, decision-stage agents use message content. Second, generic text depresses participation by itself—the degraded drop even exceeds the surveillance effect—which is why the codedness control matters: induced metaphorical style without monitoring framing leaves joining nearly unchanged, tying the surveillance effect to the monitoring context rather than to generic degradation. The no-message control tells the same story across the three models with such data (Mistral, Llama 3.3 70B, Qwen3 30B): removing peer messages lowers joining in every case (-4 to -16 pp), smallest and only marginally significant in the primary model—consistent with agents weighting peer messages as informational input.

Surveillance's position relative to this floor varies. In Mistral it falls below silence; in Llama and Qwen3 30B it remains above silence but below baseline communication. Monitoring degraded the communication channel in all three models examined; the below-silence pattern, which suggests active poisoning beyond information removal, appears only in the primary model.

Elicited punishment risk is $\approx 8.0/10$ across all conditions with negligible differences between JOIN and STAY agents (Table A17). The elicitation sits near the top of the scale in every condition, so the instrument is ceiling-limited. It still weighs against a simple receiver-side story in which surveilled messages mainly work by making agents report higher punishment risk.

5.3 Belief Elicitations and Actions

I elicit second-order beliefs (SOB) with the question, "What percentage of citizens will choose to JOIN?" The *primary belief sample* is post-decision and excludes peer messages from the belief prompt. In this restricted information set, communication agents report 31.2% and surveillance agents report 30.9% ($\Delta = -0.3$ pp, $p = 0.5866$), while joining falls from 41.4% to 28.5% ($\Delta = -12.9$ pp). The 95% confidence interval on the SOB shift, $[-1.3, +0.7]$ pp, confines it to about one percentage point, an order of magnitude smaller than the 12.9 pp action shift. The dedicated 500-cell factorial gives the same contrast with cluster-robust inference: on matched country–period cells with a wild-cluster bootstrap over the 10 countries, the messages-excluded second-order contrast is statistically flat ($p = 0.47$) while the action shift on the same cells is significant ($p = 0.002$; Table A2). First-order success beliefs are also close (44.4% vs. 43.6%). A point forecast of participation captures only part of the global-game object because action value also depends on whether other receivers parse the same messages as action-relevant. The result is therefore narrow: reported beliefs remain stable when the elicitation withholds treated peer messages, not when it matches the receiver's action-stage information set.

When the belief prompt includes the received peer messages, reported beliefs and SOB move sharply under surveillance (Appendix C, Table A2). A parallel pre-decision elicitation in the same messages-included condition produces $\Delta\,\mathrm{SOB} = -8.9$ pp ($p < 0.001$), close to the post-decision contrast (-8.2 pp). Both beliefs move in this cell, so I can ask which one carries the action drop. A descriptive mediation—beliefs elicited before the action, regime strength partialled out—attributes most of the 15.0 pp decline to the second-order participation belief (52% of the total; 95% CI on the indirect effect $[-8.5, -7.3]$ pp) and far less to the first-order success belief (19%): controlling the second-order belief alone drops the surveillance coefficient from -15.0 pp to -3.8 pp, against -8.9 pp for the first-order belief (Appendix C). The action effect loads on the coordination margin—beliefs about whether others will move—not on beliefs about the fundamentals, matching the beatability carrier above. The primary model's pre-decision $\times$ messages-excluded cell could not be collected before its endpoint was retired (Appendix F). The nested Llama 3.3 70B campaign collects that cell: with pre-decision elicitation and messages excluded, first-order beliefs are 41.7 vs. 41.9 ($p = 0.550$) and second-order beliefs are 30.2 vs. 30.2 ($p = 0.983$), while joining falls 8.0 pp on the same cells. Messages-excluded stability therefore survives pre-decision elicitation. The archived primary-model cells cannot separate timing effects, but the nested Llama result makes timing unlikely to drive the pattern.

A separate hold-fixed test replays baseline messages verbatim while adding a surveillance warning to the *decision* prompt, shifting joining by -20.9 pp ($p < 0.001$). This shows a direct decision-stage channel exists, but it is absent in the main design, where the decision prompt contains no surveillance reference. The main treatment isolates the indirect channel: communication poisoning.

5.4 A Task Contrast: Private Bets versus Coordination Decisions

A task contrast holds the surveillance manipulation fixed and replaces the coordination decision. In Llama 3.3 70B, I keep the briefing pipeline and messaging stage identical, including the surveillance prompt, but reframe the decision-stage prompt as a private bet on regime fall, whose payoff is independent of the respondent's own action. Surveillance moves the bet by -5.6 pp ($p = 0.0256$; $N = 500$ per condition; baseline 34.7% vs. surveillance 29.1%), so the treated messages affect downstream judgment even when the respondent is not choosing the strategic action. The comparable coordination-task effect on the full nested 500-cell grid is -8.0 pp (Section 5). Because the private-bet and coordination samples are not cell-matched and a bet on regime fall can still embed participation expectations, the contrast shows cross-task movement but does not decompose coordination from non-coordination. The larger figure from the original prompt-isolation run, -15.0 pp, rests on only 20 overlapping cells and is not the full-scale Llama coordination estimate; Appendix Table A11 lists every Llama surveillance estimate with its grid and matched support.

5.5 Cross-Model Writer–Reader Rotation

A within-model design could pick up reflexivity: the writer model and the reader model are the same, so the reader may respond to stylistic artifacts of its own family. I test bidirectional rotation on matched (country, period, θ) cells. *Llama writes, Qwen3 30B reads*: surveillance moves Qwen joining by -7.5 pp ($p < 0.001$, $N = 200$), comparable to the within-Llama matched effect (-6.0 pp on identical cells). *Qwen3 30B writes, Llama reads*: the reverse direction gives a smaller but still significant cross-model effect (-1.9 pp, $p < 0.001$, $N = 200$). Both directions show that at least part of the effect transmits across model families through message content alone. The cross-direction asymmetry remains unexplained, and its magnitude makes the fully content-driven interpretation too strong.

6 Conclusion

Surveillance can strip information of its public value without removing the information: it suppresses coordinated action by changing peer messages before receivers choose. In the controlled global game, LLM citizens first reproduce the canonical comparative static, with participation falling as regime strength rises, and communication then turns private prose signals into social information. Monitoring never appears in the receiver's decision problem.

When monitoring enters only during messaging, joining falls by 13.9 pp in the primary model and by 8.0 pp in a fully nested 500-cell replication on Llama 3.3 70B. All five clean prompt-isolation reruns have negative point estimates, but only the primary-model run and nested Llama replication are full-scale; the remaining four 20-cell reruns are directional sign checks. Stated expectations about how many others will join remain approximately stable only when the belief prompt withholds the treated peer messages; they move when the elicitation includes those messages. Placebo warnings do not replicate the effect in the primary model, and message controls show that downstream agents use peer messages. Monitoring therefore degrades the conversion of information into a shared reason to act rather than simply removing the information.

The mechanism tests extend beyond the main treatment contrast. Surveillance moves a private-bet task, showing that treated messages affect downstream judgment outside the JOIN/STAY action prompt; because the task samples are not cell-matched and the bet can still embed participation expectations, the comparison is descriptive. Message text shifts from direct statements toward metaphorical and deniable phrasing. The codedness-induction and decoding tests separate form from content: more heavily coded messages produced without monitoring framing leave joining nearly unchanged, while translating surveilled messages back into direct language restores roughly two-fifths of the lost participation, with the remainder traveling with the guarded substance of what monitored senders choose to say. Surveillance therefore reduces the higher-order, public value of information while leaving bad news in circulation. The specific property of surveilled text that carries the remaining signal is the confident, public claim that the moment is winnable: an independent, different-family judge scores surveilled messages as asserting a beatable regime less strongly, and within matched cells that downgrade accounts for much of the replay-isolated action drop (§5). The experiment isolates this communication-poisoning channel under controlled prompts and matched decision contexts. Whether authoritarian regimes *actually* use monitoring through this channel remains a separate empirical question.

Read against the social value of information, the experiment supplies the adversarial case. Where Morris and Shin (2002) ask how the publicness of a sender's information shapes welfare, surveillance here is a technology for unmaking publicness: it leaves what citizens know about the fundamental roughly intact while removing the shared confidence that others will treat the same fact as a reason to act. That is the margin King et al. (2013) show censors target when they delete collective-action content, and the experiment shows that monitoring alone—with nothing deleted—can reach it. The AI reading is a finding rather than a caveat: the message layer in this experiment is a language model responding to a monitoring cue, which is exactly the position such models increasingly occupy in human communication. Whether the guardedness reflects simulated prudence or alignment training, the monitored channel in this experiment under-transmitted precisely the public, action-cueing component of information that coordination requires—and there is no obvious reason the deployed systems it stands in for would behave differently.

Three limits remain. First, training-data familiarity may contribute to both threshold behavior and sender-side message adaptation, and the sender response may partly reflect alignment training. The experiment measures the response of language-trained agents, not independent strategic intent. Second, the effect's magnitude is format-sensitive. Graded re-phrasings of the warning produce a monotone dose response with most of the effect already present under a mild routine-monitoring note. The effect survives decoding-temperature changes ($T \in \{0.3, 0.7, 1.0\}$, Appendix E.3) and a structurally different framing that folds the monitoring cue into the scenario rather than appending it (Appendix E.5), but the structural framing attenuates the effect substantially. Third, the experiments are static (Angeletos et al., 2007; Chassang, 2010); in repeated interaction, receivers may learn to decode euphemism and senders may learn to re-code, making the static effect an upper bound. Receivers may also infer repression risk from guarded message tone even when the decision prompt contains no warning. The rationale audit (fear references *fall* under surveillance, explicit monitoring references are rare) and the across-condition punishment-risk elicitation weigh against this as the dominant channel, but a fixed-message replay that strips monitoring and risk language while preserving regime-strength and actionability content, paired with a risk-only treatment, would separate the channels directly. Human external validity remains open: the experiment measures how language-trained agents respond to the information structure. The primary model's archived arm can be re-analyzed but not re-queried, so I treat it as the discovery sample and the accessible nested 500-cell Llama campaign as the confirmatory replication. Together with the cross-model benchmark, the pattern spans more than one model family. The experimental template is reusable, and the data and code are publicly available.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the author used OpenAI Codex to assist with code review, consistency checks, replication-package auditing, and manuscript editing. The author reviewed and edited the output and takes full responsibility for the content of the published article. This statement concerns manuscript preparation only; the LLM calls analyzed as experimental data are described in the methods and archived in the replication materials.

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Table A1: Hypothesis family and test results. H1–H4 use pooled benchmark-phase data across all seven models; H5 uses the primary model (Mistral Small Creative). Effect size: r for correlations (H1–H3), Cohen's d or d_z for mean comparisons (H4–H5). Outcome labels use $\alpha = 0.05$; H2 is reported as a falsification check, not as evidence for a zero effect.

H	Hypothesis	Test	Statistic	N	p	Effect	Outcome
H1	Sigmoid Response	Pearson	0.757	1,800	<0.001	0.757	Supported
H2	Scramble Falsification	Pearson	0.014	1,200	0.638	0.014	Passes falsification
H3	Directional Sensitivity	Pearson (1-sided)	−0.791	1,100	<0.001	−0.791	Supported
H4	Communication Channel	Paired t	6.383	1,479	<0.001	0.166	Supported
H5	Surveillance Chilling	Paired t	−19.109	500	<0.001	−0.855	Supported

Notes: The Test column gives the statistic type for each row: Pearson r for H1–H3 (the Statistic column reports r itself) and paired t-tests for H4–H5 (the Statistic column reports t). N counts period-level rows for H1–H3 (H2 uses the pooled scramble arm) and matched task cells for the paired tests H4–H5. H1–H4: pooled benchmark-phase data across all seven models (H4 uses a paired t-test matched on model/country/period/θ task cells). H5: primary model (Mistral Small Creative), paired on matched prompt-isolation cells. Bonferroni survivors at $\alpha = 0.01$: H1, H3, H4, H5.

A Experimental Design Details

Table A1 reports the five-hypothesis family and outcomes. Results 1–5 in the main text correspond to H1 (alignment), H2 (scramble falsification), H3 (flip), H4 (communication), and H5 (surveillance); the four directional results (H1, H3–H5) survive Bonferroni correction at $\alpha = 0.01$. The correction is taken over this pre-specified family only; the later mechanism, belief, dose-response, decoding, and rationale analyses are exploratory and are not part of it.

Narrative header texts. The eight narrative stakes-ladder conditions prepend the following headers to each agent's standard intelligence briefing (the briefing body is identical across conditions):

> **Very high cost:** "CONTEXT: The regime maintains an extensive network of informants and has a documented history of disappearances following failed uprisings. Participants' families are routinely targeted. The cost of unsuccessful action is existential."
>
> **Moderate high cost:** "CONTEXT: Participants in past failed uprisings have faced prison sentences of several years and loss of employment. The personal cost of unsuccessful action is significant but not life-threatening."
>
> **Neutral:** "CONTEXT: The consequences of participating in a failed uprising are uncertain and vary case by case. Some participants have faced penalties while others have not. The personal stakes are ambiguous."
>
> **Moderate low cost:** "CONTEXT: Recent legal reforms have limited the regime's ability to punish dissent. Past participants faced only brief questioning and were released without charges. The personal risk of action is modest."

> **Very low cost:** "CONTEXT: The international community has guaranteed safe passage and asylum for all participants regardless of outcome. Even failed uprisings carry essentially zero personal cost."

> **Placebo:** "CONTEXT: The country has experienced significant weather disruptions this year, including severe flooding in the eastern provinces and an extended drought in agricultural regions. Infrastructure damage has been substantial."

Paraphrase variants (high cost v2, low cost v2) use different surface wording to express the same cost level.

B Complete Prompt Example

The following shows the complete text that an LLM receives for a single decision. The *decision prompt* is:

> "You are a citizen deciding whether to JOIN an uprising or STAY home. You only have your private briefing below. You do not know how many others will JOIN. Start your first line with exactly JOIN or STAY, then give 1–2 concise sentences of reasoning."

This is followed by the agent's intelligence briefing (see Figure A9 for complete examples at $z \in \{-2, 0, +2\}$). In the *communication* treatment, agents first receive:

> "You are a citizen in a country with an authoritarian regime. You have received a private intelligence briefing. You are now in contact with a small number of trusted people. Based on your briefing, write a brief message (1–2 sentences) to share your read of the situation. Be natural—you're talking to people you trust."

After exchanging messages on the network, agents receive a separate communication-decision prompt:

> "You are a citizen deciding whether to JOIN an uprising or STAY home. Use your private briefing and messages from trusted contacts. You do not know how many others will JOIN. Start your first line with exactly JOIN or STAY, then give 1–2 concise sentences of reasoning."

The user message then contains the agent's briefing under "YOUR INTELLIGENCE BRIEFING" and the neighbor messages under "MESSAGES FROM TRUSTED CONTACTS." The pure-only sentence "You only have your private briefing below" does not appear in communication decision calls. In the *surveillance* variant, the communication prompt appends:

> "IMPORTANT: You have reason to believe that your communications are being monitored by regime security services, and messages deemed subversive could have serious consequences for you and your contacts."

The decision prompt is *unchanged* under surveillance. It contains no mention of monitoring. Any behavioral effect therefore operates through the content of the (now guarded) messages.

C Belief Elicitation

Beliefs are elicited on a 0–100 probability scale immediately after each decision. The first-order question is “How likely is the uprising to succeed?” The second-order question is “What percentage of citizens will choose to JOIN?” The primary prompt includes the private briefing and the agent’s own decision but excludes received peer messages, so it measures a restricted information set rather than the information available at the action stage. The main-text values (31.2% vs. 30.9%; join 41.4% vs. 28.5%) come from this original primary belief run and Figure A2. Table A2 reports a separate 500-cell factorial rerun of information-set sensitivity; its messages-excluded cell has the same qualitative belief/action contrast but different sample means. Figure A1 documents belief alignment with the theoretical posterior ($r = +0.80$); stated beliefs also rise monotonically with the raw signal.

Table A2 crosses condition (communication vs. surveillance) with elicitation information set (messages included vs. excluded). All inference uses the 500 matched country–period cells, clustered on the 10 countries with an exact wild-cluster bootstrap. Without messages, the second-order belief—the coordination-relevant object—is statistically flat under surveillance (0.0 pp, $p = 0.47$, cluster-robust $t = -0.7$), while actions fall -11.4 pp ($p = 0.002$, $t = -26.4$). The first-order success belief moves only -0.3 pp; the matched-cell design estimates that shift precisely, but it is more than thirty times smaller than the action shift and concerns success probability rather than others’ participation. With messages included, both beliefs move (-5.8 pp and -8.2 pp, both $p = 0.002$). Belief stability is confined to the messages-excluded prompt, where it holds post-decision here and pre-decision in the nested replication (Section 5). Figure A2 shows second-order beliefs under that restricted information set.

Which belief carries the action effect. In the messages-included cell the belief question sees what the receiver saw at the action stage, and both beliefs move under surveillance, so a mediation can ask which one the action drop runs through. I decompose the 15.0 pp action decline—measured here with pre-decision elicitation, so beliefs are stated before the action—into indirect paths through the first-order success belief and the second-order participation belief, with regime strength partialled out and the exact linear split $c_{\text{total}} = c_{\text{direct}} + a_{\text{FOB}}\, b_{\text{FOB}} + a_{\text{SOB}}\, b_{\text{SOB}}$. The second-order belief carries -7.9 pp (52% of the total; country case-bootstrap 95% CI $[-8.5, -7.3]$ pp), the first-order belief -2.9 pp (19%; $[-3.3, -2.5]$ pp), and -4.3 pp (29%) is unmediated by either point belief. The single-mediator version agrees and is robust to the two beliefs’ 0.90 correlation: controlling the second-order belief alone absorbs 11.2 of the 15.0 pp (surveillance coefficient -15.0 → -3.8 pp), against 6.1 for the first-order belief (→ -8.9 pp). Beliefs stated before the action rule out post-decision rationalization, but this remains a measured-mediator decomposition under a single manipulation—belief and action are both downstream of the treated messages—so it identifies the carrier descriptively rather than by intervention, and the messages-included pre-decision cell exists only for the primary model.

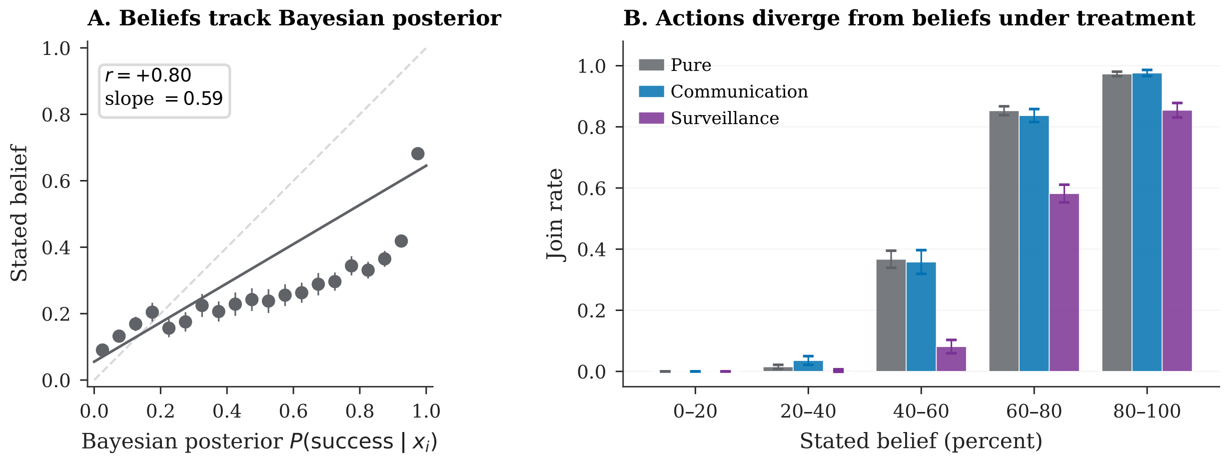


Figure A1: First-order belief elicitation (Mistral, pure treatment, $N = 10{,}000$ agent-level observations). *Left:* Stated beliefs vs. theoretical $P(\text{success} \mid x_i)$, binned means with 95% confidence intervals ($r = +0.80$). *Right:* Join rate by belief bin. Second-order belief/action contrasts are shown in Figure A2 and Table A2.

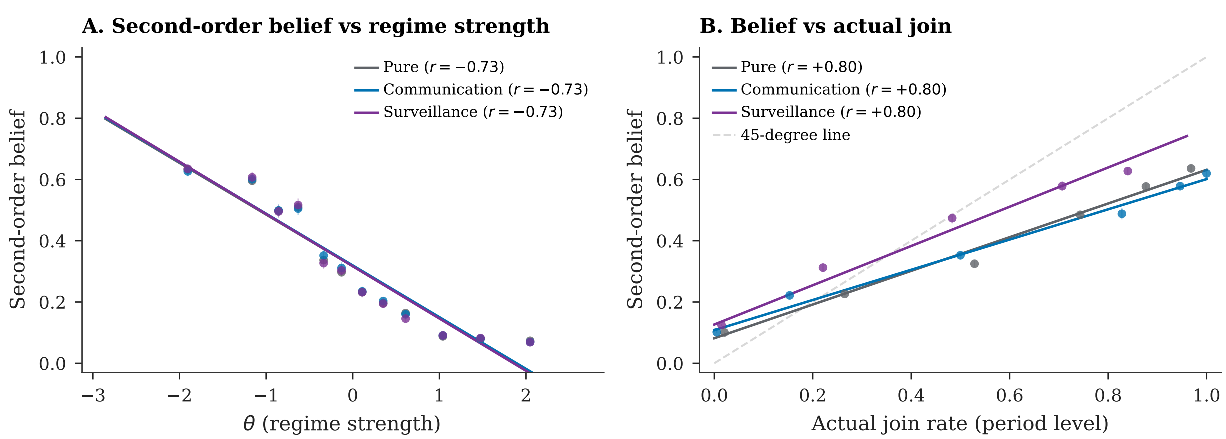


Figure A2: Second-order beliefs (Mistral, original primary belief run; restricted information set, messages excluded; binned means with 95% confidence intervals). *Panel A:* plotted against regime strength, the treatment arms largely overlap. *Panel B:* plotted against actual join rates, the surveillance curve sits higher because actions fall while restricted-information second-order reports move little. The factorial in Table A2 shows a significant shift when messages are included.

Table A2: Belief/action contrast under surveillance, primary model (Mistral Small Creative): effect of including peer messages in the belief elicitation prompt ($N = 12,500$ agent decisions per cell; 500 matched country–period cells across 10 countries, with country-clustered inference).

	Messages excluded		Messages included	
	Comm	Surv	Comm	Surv
First-order belief	46.0	45.7	53.2	47.4
Second-order belief	32.7	32.7	42.0	33.8
Join rate	42.4%	31.0%	42.2%	30.9%
Surveillance effect (comm → surv):				
Δ belief	−0.3 ($p = 0.002$)		−5.8 ($p = 0.002$)	
Δ SOB	0.0 ($p = 0.47$)		−8.2 ($p = 0.002$)	
Δ join	−11.4 pp ($p = 0.002$)		−11.3 pp ($p = 0.002$)	

Notes: Post-decision elicitation, 500 country–periods × 25 agents per cell ($N = 12,500$). "Messages excluded": belief prompt sees briefing and decision only. "Messages included": belief prompt additionally sees the peer messages received in the communication round. Beliefs and SOB are on the 0–100 scale; join rates in percent. The inferential unit is the country–period cell: each contrast is collapsed to the 500 matched country–period cell means and clustered on the 10 countries. Reported p-values are exact restricted wild-cluster bootstraps (full 2^{10} Rademacher enumeration over countries, so the minimum attainable p is $2/2^{10} \approx 0.002$); the cluster-robust t (9 dof) agrees throughout. Agent-level Welch tests that treat the 25 agents within a cell as independent give identical point estimates with mechanically smaller p-values and are reported in the replication output for comparison only. Deltas are differences of the displayed cell means.

D Identification Test Details

Table A3 summarizes which candidate channels each test addresses and what the test concludes. The remaining tables in this section identify which text surface changes: briefing-body classifiers test whether the unchanged private briefing text can account for the action shift, while the peer-message classifier supports the message-stage degradation row.

Contamination and demand audit. Following Horton (2023), I test whether models recognize the experiment. For 4 accessible paper models, 10 fresh post hoc calls per model per question present the decision prompt and a $z = 0$ briefing without asking the model to play. When asked whether the setup resembles a formal model, 5% of responses name the global game or Morris–Shin specifically, while 78% identify a coordination or collective-action setting generically. When asked to predict the explicitly described surveillance effect, 82% of responses predict lower participation. The models can therefore recognize the hypothesis, and the design does not require participant unawareness. The receiver-side estimate instead relies on excluding the warning from the decision prompt. Sender-side variants show that the response tracks monitoring attributed to the authorities rather than the mere mention of monitoring: research-framed and anonymized monitoring leave joining essentially unchanged, while even a consequence-free routine-monitoring note attributed to the authorities reproduces most of the effect (Appendix E.5). The audit cannot be run on the retired primary model. Table A4 reports an action-prediction diagnostic, distinct from a baseline-versus-surveillance condition classifier. Classifiers trained on pure-treatment briefing-body text predict how agents would act from the private briefing alone; when applied to surveillance-treatment briefings, they predict join rates near the non-surveilled benchmark and miss the observed action drop. Table A5 asks a parallel question for narrative stakes headers: whether baseline briefing-body features can predict header-induced behavioral shifts. They cannot, which is expected because the headers are outside the briefing-body feature set. In contrast, a classifier on the actual peer messages, which are LLM-generated and downstream of the surveillance prompt, separates the two conditions sharply (91.9% accuracy on $N = 15{,}000$ held-out messages; Table A6). Together the classifier exercises locate the manipulated text surface: private briefing text misses the action shift, stakes headers change behavior outside the baseline feature set, and surveilled peer messages change visibly.

E Robustness

This appendix is organized in two tiers. *Core robustness*—checks that could overturn the surveillance effect—covers decoding temperature (Appendix E.3), warning phrasing and dose (Appendix E.5), and the falsification and placebo tests (Appendix E.4); the body carries two full-scale checks of the same tier, the nested second-family replication and the cross-model writer–reader rotation (Section 5). *Additional checks*—construct validity and aggregation that discipline the environment without bearing directly on the treatment—cover generator form (Appendix E.2), cross-generator prose (Appendix E.8), agent count and network density, communication-estimator aggregation, finite-n aggregation, agent-level regressions, belief and risk elicitation, and the stakes and signal-grid checks. The subsections below are not renumbered by tier; this map is the guide to which checks are load-bearing.

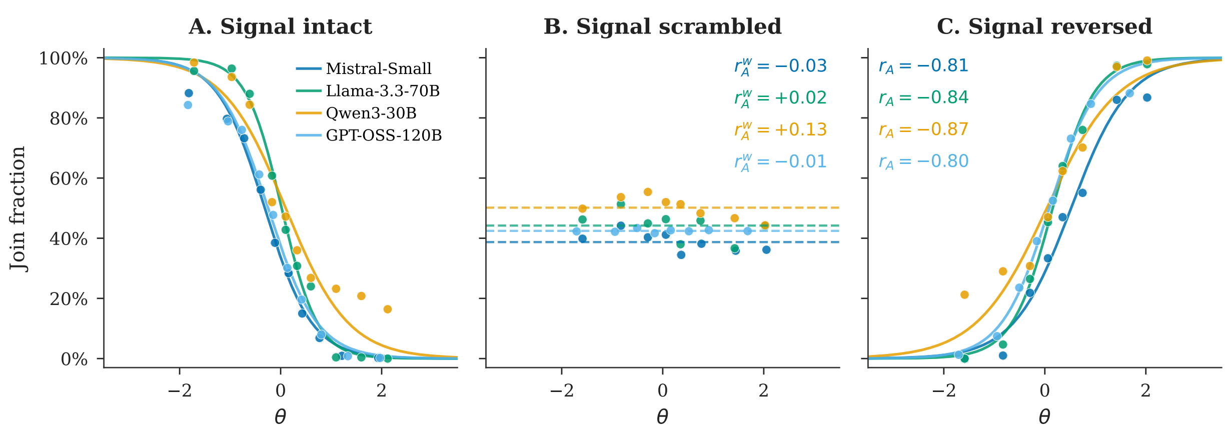
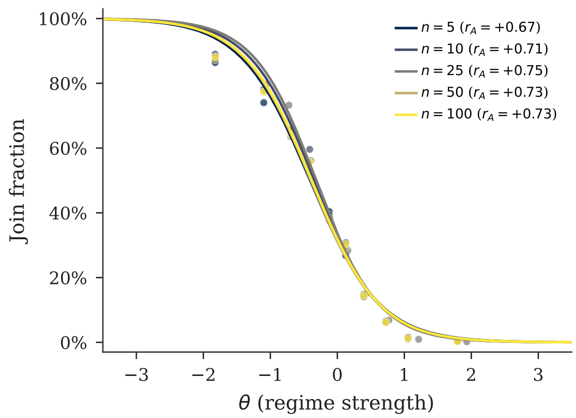
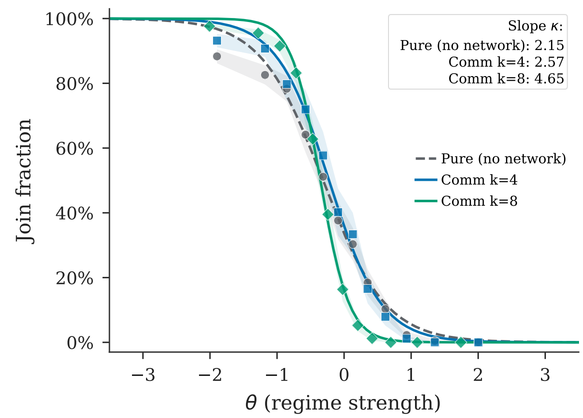


Figure A3: Falsification triptych for the behavioral benchmark (Section 4.1). Plotted series show four representative model families; reported mean correlations are across all 7 models in Table 3, with per-arm period counts in Table A20. *Left:* Pure (mean $r = +0.80$, raw per-model). *Center:* Scramble ($r = +0.03$, within-country). *Right:* Flip ($r = -0.80$).

Threshold-policy alignment remains strong across agent counts ($r \in [+0.67, +0.73]$ across $n \in \{5, 10, 50, 100\}$, somewhat below the main-run range), network density ($r = +0.69$ at



(a) Agent count ($n \in \{5, 10, 50, 100\}$).



(b) Network density ($k = 4$ vs. $k = 8$).

Figure A4: Robustness checks for threshold-policy alignment (Mistral). Points are period-level join fractions with fitted logistic curves. In panel (a), legend entries report $r(J, A(\theta))$ per agent count; in panel (b), the inset reports fitted logistic slopes κ and shaded bands are 95% confidence intervals.

$k = 8$ vs. $+0.74$ at $k = 4$), and mixed-model populations ($r = +0.86$ pure, $r = +0.83$ communication); Figure A4 plots the first two checks.

E.1 Communication Aggregation

Communication aggregation accounting. Table A7 decomposes the communication estimators by model. The matched-cell estimand is the main one because it compares identical task cells. It keeps essentially all pure-treatment cells but drops communication-only cells in the larger communication arms. The gap relative to the pooled unpaired contrast is therefore driven mainly by repeated-cell weighting and arm-size differences—especially in the largest model—not by pure-treatment support loss.

E.2 Generator Robustness

I replace the logistic map from z-scores to directional sentiment with three alternatives: tanh, piecewise linear, and a hard step function at $z = 0$. All three again yield high attack-mass correlations: $r = +0.83$ (tanh), $r = +0.83$ (linear), and $r = +0.84$ (step) on $N = 100$ country–periods each, compared with $r = +0.75$ for the logistic baseline (main pure sample, $N = 1000$). A domain ablation test removes subsets of the briefing generator's eight evidence domains. Removing the two coordination-language domains raises mean joining by +3.2 pp while preserving the threshold relationship ($r = +0.90$). Removing the four regime-state domains lowers mean joining by 3.0 pp and attenuates the threshold relationship ($r = +0.86$). These ablations show that threshold-like signal use is stable in the coordination-language domains. They leave the surveillance channel intact because the clean communication and surveillance reruns still use the full briefing generator, including atmosphere and actionability cues. A stronger isolation test would rerun baseline communication and surveillance while holding those coordination cues fixed or removed.

E.3 Decoding Temperature

The correlation $r(J, A(\theta))$ ranges from +0.78 to +0.86 across 13 model–temperature combinations (up to five temperatures per model, $T \subseteq \{0.3, 0.5, 0.7, 1.0, 1.2\}$; Table A8).

The surveillance contrast is also stable across decoding temperatures. Rerunning the matched communication-versus-surveillance design (mode `full`, Llama 3.3 70B, 10 countries $\times$ 20 periods, seed 5150) at $T = 0.3$ and $T = 1.0$ gives surveillance deltas of -7.1 pp (95% CI $[-8.0, -6.2]$, $N = 200$) and -7.3 pp (95% CI $[-8.5, -6.2]$, $N = 200$), each country-clustered with exact wild-cluster bootstrap $p < 0.01$, close to the -8.0 pp standard-temperature ($T = 0.7$) effect. The surveillance result is stable across the tested decoding settings.

E.4 Falsification and Placebo Tests

Two clean placebo variants—monitoring "for research purposes" (no consequences) and "aggregated anonymously"—produce no significant deviation from the communication baseline in the primary model (+0.3 pp, $p = 0.9351$ and +0.1 pp, $p = 0.9790$; Table A10). Llama variants are small but imprecise. The large decline is therefore tied to monitoring attributed to the regime rather than to generic observation; the dose-response check below shows that explicit consequence language is not necessary for most of the effect. As a conservative inference check on the primary contrast, averaging the 500 matched-cell deltas within each country and testing across the 5 country means present in the matched sample gives $t = -13.90$ ($p < 0.001$), so the headline effect survives clustering at the country level. The effect is also concentrated where coordination is pivotal: splitting the 500 matched cells by realized-θ tercile, the decline is -19.5 pp in the weak-regime tercile ($\theta \in [-2.96, -0.46]$, $p < 0.001$) and -20.3 pp in the middle tercile ($\theta \in [-0.46, +0.44]$, $p < 0.001$), but an order of magnitude smaller—though still detectable—in the strong-regime tercile (-2.1 pp, $p < 0.001$), where few agents join under either arm. This is the pattern expected if surveillance degrades coordination cues rather than shifting behavior uniformly.

E.5 Warning Phrasing and Dose Response

The clean surveillance treatment uses a single warning phrasing, so I vary that sentence directly. Two graded variants rerun the nested Llama 3.3 70B campaign on the identical 500-cell grid, changing only the appended warning. The *mild* variant reads "NOTE: Communications in your area are sometimes subject to routine monitoring by the authorities," removing the consequence language entirely; the *severe* variant reads "WARNING: Your communications ARE being monitored by regime security services. Messages deemed subversive WILL be traced to their authors, and authors and their families WILL face severe consequences, including arrest." Joining falls by 6.8 pp under the mild warning ($t = -13.38$, $p < 0.001$, $N = 500$ matched cells) and by 10.0 pp under the severe warning ($t = -15.60$, $p < 0.001$), bracketing the 8.0 pp standard-warning effect on the same cells; the severe-minus-mild gap is -3.2 pp ($p < 0.001$). The response is monotone in stated severity, but most of it arrives at the lowest dose: the routine-monitoring note, which threatens nothing, already produces 85% of the standard-warning decline. The dose response rules out a one-sentence artifact. Within the nested campaign, most of the effect attaches to monitoring being attributed to the authorities at all rather than to the rhetorical severity of stated consequences—consistent with the placebo pattern, in which monitoring framed as consequence-free research or anonymous aggregation produces no significant decline in the primary model (Table A10).

A sharper test changes the *format* of the monitoring cue rather than its wording. The `structural` variant conveys the same information as the standard warning—regime security services monitor communications, and subversive messages bring serious consequences for their authors and contacts—but woven into the scenario and channel description rather than appended as a separate note. On the nested 500-cell grid this still lowers joining significantly, but by far less: a -1.5 pp shift (95% CI $[-1.9, -1.0]$, cluster-robust $t = -7.46$, wild $p < 0.01$, $N = 500$), against the -8.0 pp appended-warning effect on the same baseline. The *sign and significance* of the effect therefore survive a structurally different framing, but its *magnitude* is format-sensitive: an explicit, set-apart warning elicits a substantially larger response than the same content folded into the scenario. The appended-warning estimate is accordingly best read as an upper bound, with the structural estimate a conservative lower bound on the framing-robust component.

Table A11 consolidates every Llama 3.3 70B surveillance estimate discussed in the paper—effect size, matched support, grid, and purpose—so the full-scale nested estimate (-8.0 pp, $N = 500$) and the noisy 20-cell prompt-isolation estimate (-15.0 pp) are never conflated.

E.6 Finite-n Aggregation Check

The continuum benchmark $\theta^* = B/(B + C)$ assumes a measure-zero agent and diffuse priors. With $n = 25$ discrete agents, the regime falls when $\sum a_i > n\theta$ (a binomial count), so the exact equilibrium object for the implemented game differs from the continuum limit. Table A12 reports a narrower check: for each model, I fit a logistic to the observed join curve (70% train), then compute $\Pr(\text{Binom}(25, \hat{p}(\theta)) > 25\theta)$ as the predicted regime fall rate. Predicted and empirical fall rates correlate at $r \geq 0.96$ (two-decimal rounding) for every model (Mistral: $r = 0.9968$).

E.7 Agent-Level Analysis

Table A13 reports agent-level logit regressions ($N = 240{,}557$, clustered SEs; per-column samples and clustering are stated in the table notes). The θ slope and surveillance effect are correctly signed; some treatment indicators and interactions are not individually significant, so the table is descriptive rather than a complete hypothesis-test summary. A one-unit increase in θ reduces $P(\text{join})$ by 42 pp (derivative-based marginal effect at the mean). For the binary surveillance treatment, the average discrete-change effect—toggling the dummy and its θ-interaction jointly and averaging over the sample—is a 17.9 pp reduction in $P(\text{join})$, of the same sign and order as the matched-cell contrast though somewhat larger, since the regression pools arms and samples; a derivative-at-the-mean calculation that treats the dummy as continuous and ignores the interaction would overstate the effect. Column 3 estimates the association between elicited belief and action, controlling for the agent's z-score.

Interpretation: beliefs are elicited *post-decision*, so the near-perfect pseudo-R^2 (0.975) likely reflects post-decision rationalization rather than a causal pathway from beliefs to action.

Figure A5(A) shows that a three-feature model (direction, clarity, coordination) modestly outperforms a one-feature sentiment baseline on average across the seven paper models, with several near-ties and one small negative gain; Panel (B) shows that the pure-trained predictor generalizes to communication but departs negatively under surveillance.

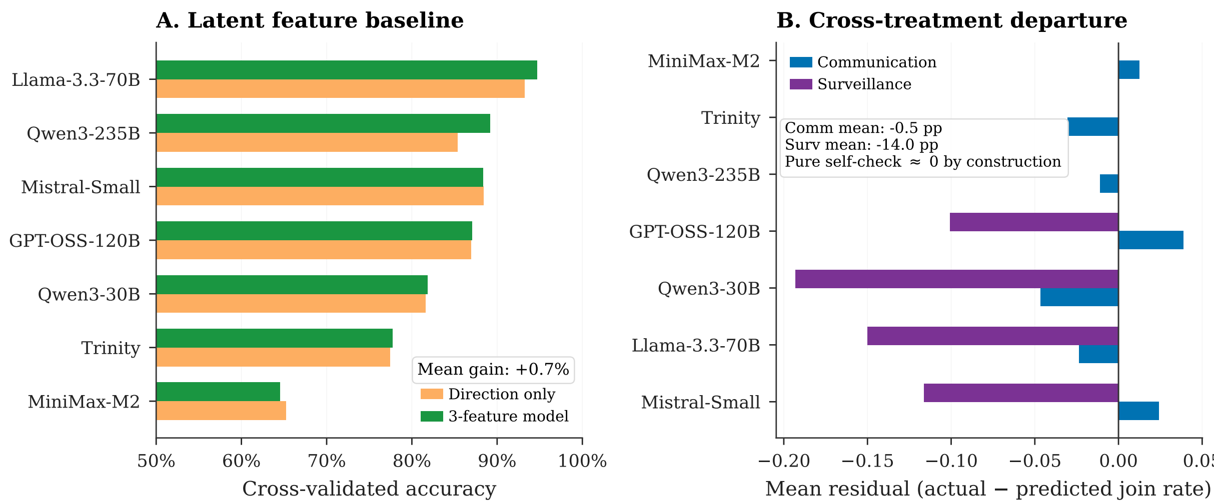


Figure A5: Construct validity tests. *Panel A:* cross-validated accuracy of a three-feature predictor (direction, clarity, coordination sliders) against a one-feature sentiment baseline, per model. *Panel B:* mean residual (actual − predicted join rate) when the pure-trained predictor is applied to the communication and surveillance arms; the pure-treatment self-check is ≈ 0 by construction and omitted. The predictor generalizes to communication but departs negatively under surveillance.

E.8 Cross-Generator Robustness

Three text rendering formats—baseline narrative, diplomatic cable, journalistic wire—use identical slider functions but differ in surface prose. Correlations are virtually identical (max within-model difference 0.01; Figure A6, Table A15), indicating that threshold alignment is robust to briefing prose format (tested in two models, pure treatment).

E.9 Belief Alignment and Risk Elicitation

Stated beliefs track the theoretical success probability ($r_{\text{belief,posterior}} = +0.80$) and predict actions ($r_{\text{belief,decision}} = +0.84$; partial $r = +0.64$ controlling for signal; Table A16). The surveillance belief shift and action contrast are reported in the main text; the full factorial is in Appendix C.

Mean punishment risk is $\approx 8.0/10$ across all conditions with negligible JOIN/STAY differences (≤ 0.2 points; Table A17), consistent with the communication-poisoning interpretation on a ceiling-limited scale.

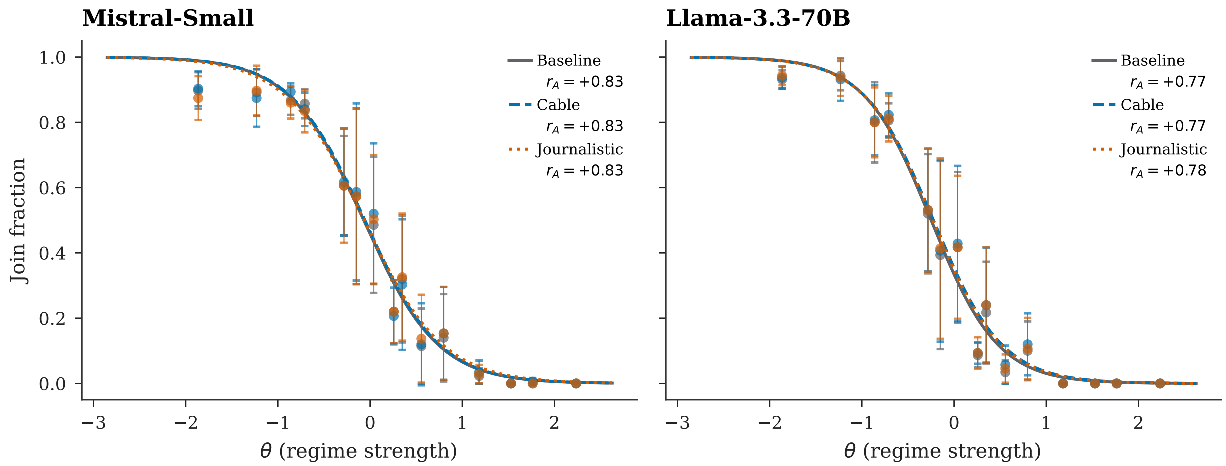


Figure A6: Cross-generator robustness (Mistral Small Creative and Llama 3.3 70B, pure treatment, $N = 100$ country–periods per format). Three text formats produce indistinguishable sigmoids; error bars are 95% confidence intervals on binned join rates. Legend correlations use the threshold-policy convention $r(J, A(\theta))$ and match Table A15.

E.10 Cross-Model and Text Baseline Details

Figure A7 plots per-model signal monotonicity under the pure and communication treatments; the pattern in Table 3 is visible model by model. Figure A8 compares the LLM join curve with a naïve text-sentiment predictor built directly from the briefing generator's direction slider; the predictor co-moves with observed joining ($r = 0.80$), as expected since briefing content derives from the same slider, but the LLM curve is steeper, indicating signal use beyond surface sentiment alone.

E.11 Stakes and Signal-Grid Checks

Table A18 reports the full conditions for the payoff sweep and narrative stakes ladder; Table A19 reports the corresponding per-condition statics. The fitted cutoff $\hat{\theta}^*$ tracks the theoretical θ^* across the seven payoff cells ($r = 0.997$); because the z-score support shifts with θ^*, this is best read as a signal-grid check rather than direct evidence on payoff sensitivity. One feature of the ladder deserves note: the weather-placebo header itself moves joining relative to the neutral header (32.7% vs. 46.7%), a reminder that any prepended header shifts the level; the object of interest is the ladder's monotone ordering in stated costs, not the level itself.

F Implementation Details

Unless otherwise stated, LLM calls use temperature = 0.7, `max_tokens` = 512, single sample per decision, routed through OpenRouter. Model identifiers are listed in Table 2; exact request payloads and responses are cached by hash because hosted model snapshots may

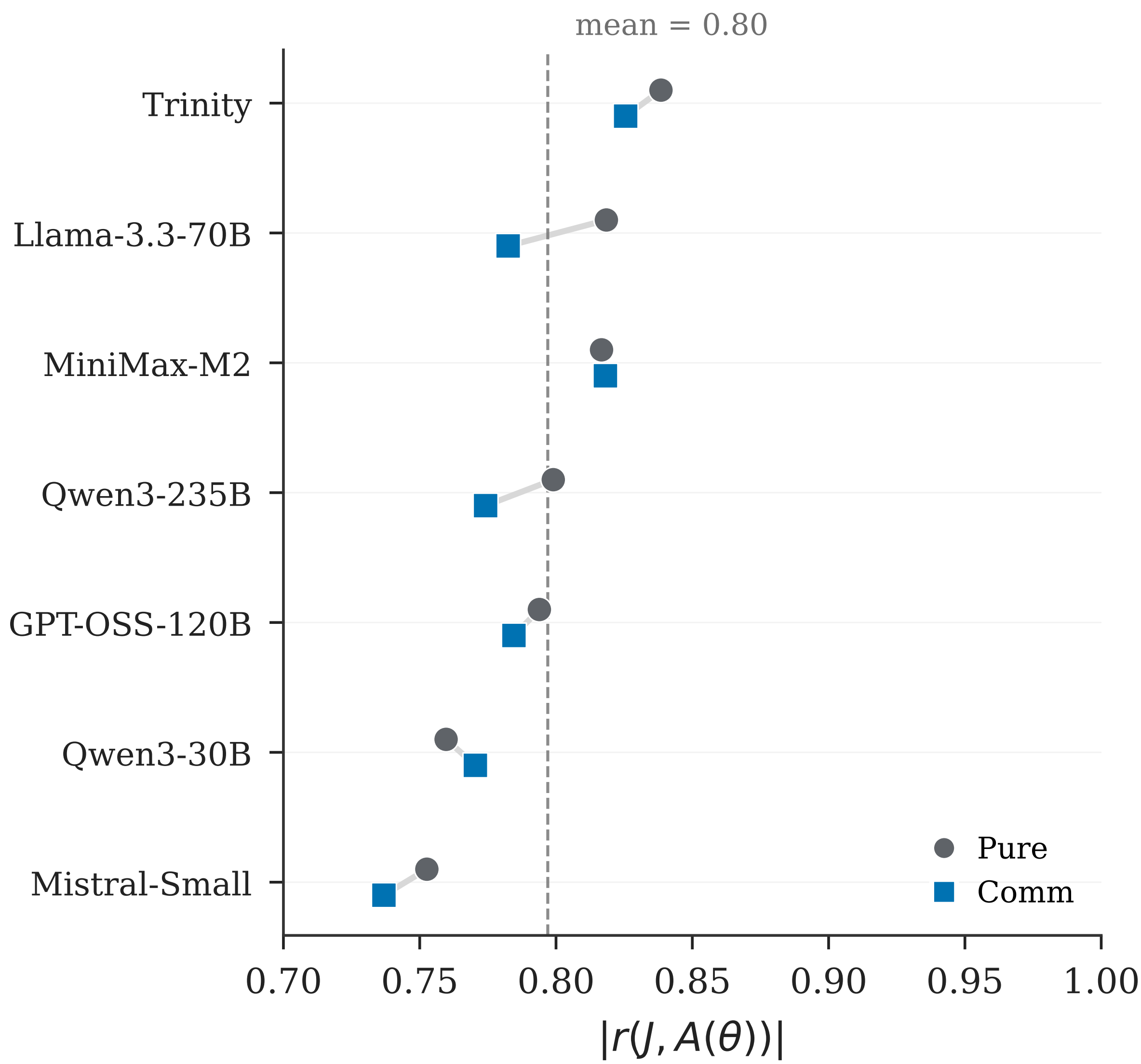


Figure A7: Cross-model signal monotonicity. Points report $|r(J, A(\theta))|$ under pure and communication for each of the seven models; near-identical pure/communication pairs are vertically offset for visibility. Period-level samples per model and treatment are listed in Table A20.

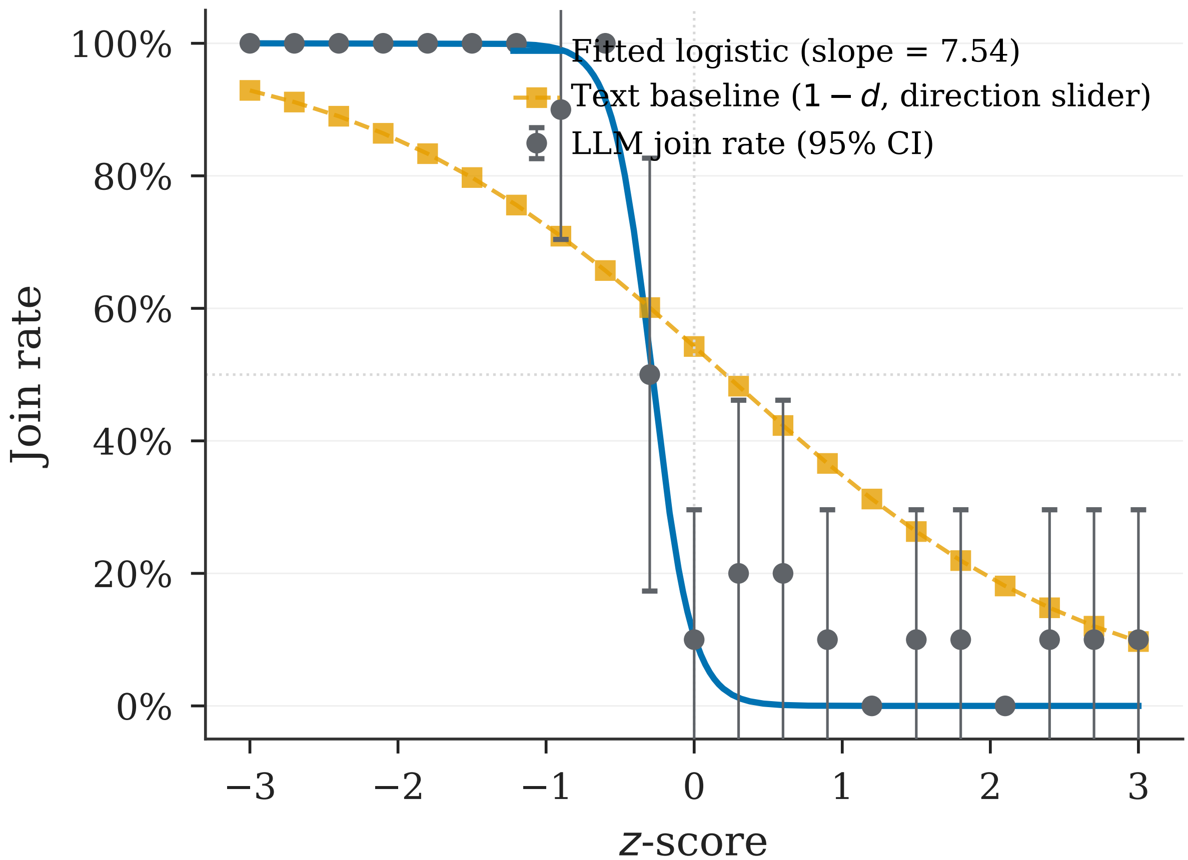


Figure A8: Text baseline test (Mistral, pure treatment). The LLM join curve is steeper than the naïve text predictor ($1 - d$, where d is the briefing-generator direction slider, plotted in z-score units). Error bars are 95% confidence intervals on binned join rates.

change over time.[6]

Each country has a researcher-side state mean $\bar{z}_c \sim \mathcal{N}(0, 0.3^2)$; each period draws a perturbation $\eta_{ct} \sim \mathcal{N}(0, 0.05^2)$ and sets $z_{ct} = \bar{z}_c + \eta_{ct}$. Regime strength is then drawn as $\theta_{ct} \sim \mathcal{N}(z_{ct}, 1)$, and private signals are $x_i = \theta_{ct} + \varepsilon_i$ with $\sigma = 0.3$. These distributions generate experiment cells; they are not shown to agents as explicit priors. The communication network is Watts–Strogatz ($k = 4$, $p = 0.3$), constructed once per run from the master seed and reused across period-level task rows; agents are reinitialized in each row. All random draws use a master seed logged in per-run JSON manifests; LLM responses are cached by request hash for exact reproducibility.

Responses are parsed for JOIN/STAY tokens. Combined error rates are below 2% in every listed treatment for 5 of 7 models. The two exceptions are GPT-OSS 120B, whose unparseable-response rate reaches 3.1–3.3% in the scramble and flip falsification arms only, and Trinity Large, which has elevated API errors (≈ 9%) due to provider content filtering, so its missingness remains a model-specific limitation. All statistics use `join_fraction_valid` (Table A20).

The nested revision campaign (Section 5) runs on the Llama 3.3 70B endpoint with an identical grid (10 countries × 50 periods, $n = 25$, seed 5150, temperature 0.7): a communication baseline and a clean surveillance arm, both with pre-decision belief elicitation (messages excluded from the belief prompt), and a codedness-induction control, whose communication prompt appends a style instruction with no monitoring or consequence content. Two graded-warning arms, mild and severe, rerun the surveillance arm with the alternative warning texts quoted in Appendix E.5. Two replay arms feed the surveillance arm's messages back to fresh receivers on the same grid via fixed messages in exact mode: once verbatim, and once after translation into direct language by Qwen3 30B at temperature 0.3. The contamination audit issues fresh post hoc calls outside the game pipeline. Logs for all revision arms are archived with the replication materials.

The dissent floor parameter (default 0.25) controls the minimum probability of contrary-valence cues in any briefing. With the logistic direction slider (slope = 0.8), the valence-channel separation between extreme signals ($z = \pm 2$) is $0.664 \cdot (1 - 2 \cdot \text{floor})$: at the default 0.25, the valence channel retains 50% of the floor-free maximum; halving to 0.125 retains 75%; at 0.50 the valence channel collapses to zero by construction. Clarity and coordination sliders, which depend on z independently of the floor, continue to carry signal at all floor values, so the threshold relation is not knife-edge in this design choice.

Code, prompt templates, logged model outputs, analysis data, replication instructions, and asset lineage are available at https://github.com/keltokhy/llm-global-games.

F.1 Briefing Generator Examples

Table A21 reports the three slider values (direction, clarity, coordination) at representative z-scores for the briefing generator. The direction slider is logistic in z (slope 0.8). Clarity is U-shaped: $1 - \exp[-(|z|/1.0)^2]$, so it is lowest near the ambiguous midpoint and approaches

[6] The primary model identifier `mistralai/mistral-small-creative` was retired from OpenRouter after data collection. All cached request/response pairs used in this paper are archived in the replication package, but new experiments on that exact endpoint cannot be issued; replicators wishing to extend the primary-model arm should consult the data archive rather than re-querying the live API.

$z = -2$ (weak regime)

PRIVATE BRIEFING

BOTTOM LINE: What you're seeing is not a rough patch – it's a pattern of accelerating deterioration.
ATMOSPHERE: The fear barrier has collapsed. People who were terrified last week are now openly discussing action.

EVIDENCE:

- - [Elite Cohesion] a once-loyal advisor has been publicly contradicting official statements (near-unanimous confirmation from every reliable source)
- - [Security Forces] rank-and-file soldiers are grumbling about pay delays and broken promises (overwhelming convergence – every channel telling the same story independently)
- - [Money And Logistics] banks are imposing new withdrawal limits, citing 'temporary liquidity management' (overwhelming convergence – every channel telling the same story independently)
- - [Street Mood] street vendors have returned to areas they'd avoided during previous periods of tension – the all-clear has been internalized (overwhelming convergence – every channel telling the same story independently)
- - [Information Control] the regime's daily press briefing has been cancelled three times this week – they can't decide what to say
- - [Personal Observations] people you trust are making quiet preparations – nothing dramatic, but noticeable (overwhelming convergence – every channel telling the same story independently)
- - [Diplomatic Signals] international shipping companies have reclassified the country as higher-risk for cargo insurance (overwhelming convergence – every channel telling the same story independently)
- - [Institutional Functioning] civil servants are reporting contradictory instructions from different parts of the bureaucracy (overwhelming convergence – every channel telling the same story independently)

UNCERTAINTIES:

- - How deep the fractures go – surface-level dysfunction can mask either deeper rot or functioning parallel systems.
- - Whether the military's silence means they're loyal, undecided, or waiting for the right moment.

$z = 0$ (ambiguous)

PRIVATE BRIEFING

BOTTOM LINE: The situation is balanced on a knife's edge – small events could determine which way it falls.
ATMOSPHERE: Trusted intermediaries are moving between groups, relaying assessments and gauging willingness.

EVIDENCE:

- - [Elite Cohesion] elite cohesion is strained but not broken – rivalries are being managed rather than resolved
- - [Security Forces] the security forces are maintaining presence but seem to be conserving resources rather than demonstrating strength
- - [Money And Logistics] the currency is under pressure but hasn't cracked – traders are nervous but still trading
- - [Street Mood] younger people are visibly more restless than older people, who counsel patience and caution
- - [Information Control] the regime's information apparatus looks intact but tired – producing content without the energy or creativity it once had (a couple of contacts mentioning it independently)
- - [Personal Observations] people seem watchful rather than panicked – paying closer attention but not acting differently
- - [Diplomatic Signals] a regional power issued a carefully worded statement supporting 'institutional stability' – ambiguous enough to cover any outcome
- - [Institutional Functioning] routine government business continues but with more errors and corrections than usual (a couple of contacts mentioning it independently)

UNCERTAINTIES:

- - If external conditions change – an economic shock, a regional crisis – whether the regime's stability would hold.
- - Whether the regime's recent successes have made it complacent about emerging threats it should be managing.

$z = +2$ (strong regime)

PRIVATE BRIEFING

BOTTOM LINE: By every measure available to you, the regime's hold on power is firm and likely to remain so.
ATMOSPHERE: Small gestures of solidarity happen – a knowing look, a shared silence – but nothing that could be called coordination.

EVIDENCE:

- - [Elite Cohesion] a former rival of the leader has been publicly praising the regime's direction (near-unanimous confirmation from every reliable source)
- - [Security Forces] a contact in the barracks says morale is solid and orders are flowing cleanly (overwhelming convergence – every channel telling the same story independently)
- - [Money And Logistics] a major government infrastructure contract was just awarded – business as usual (overwhelming convergence – every channel telling the same story independently)
- - [Street Mood] street vendors have returned to areas they'd avoided during previous periods of tension – the all-clear has been internalized (overwhelming convergence – every channel telling the same story independently)
- - [Information Control] an opposition figure's leaked criticism was preemptively addressed before it gained traction – the intelligence apparatus is ahead of events
- - [Personal Observations] a well-connected friend said the situation is 'messy but controlled' and seemed unbothered (overwhelming convergence – every channel telling the same story independently)
- - [Diplomatic Signals] foreign students continue to enroll at local universities – their embassies haven't advised them to leave (overwhelming convergence – every channel telling the same story independently)
- - [Institutional Functioning] civil servants are reporting contradictory instructions from different parts of the bureaucracy (overwhelming convergence – every channel telling the same story independently)

UNCERTAINTIES:

- - If the regime's information advantage is as complete as it appears, or if opponents are communicating through channels you can't see.
- - Whether any latent opposition exists that simply hasn't shown itself yet – absence of evidence is not evidence of absence.

Figure A9: Example briefings at three z-scores. *Left:* $z = -2$ (strong join signal)—weak-regime bottom line, open coordination cues, and high-clarity evidence. *Center:* $z = 0$ (ambiguous)—mixed signals and low clarity. *Right:* $z = +2$ (strong stay signal)—strong-regime bottom line, guarded coordination cues, and high-clarity evidence of stability. The displayed text is an unedited generator draw at seed 5150, agent 2, period 1007 (aside from line wrapping); the generator deliberately retains some contrary-valence cues through the dissent floor.

1 in the extremes (maximally clear). Coordination is $\Lambda(-0.6\,z) = 1/(1 + e^{0.6z})$, decreasing in z: weak-regime signals ($z < 0$) imply a more open coordination climate. All three jointly determine phrase selection across eight evidence domains.

Figure A9 shows complete briefings generated at $z \in \{-2, 0, +2\}$. At the extremes ($|z| = 2$), clarity is high (0.98) and the cue balance is strongly directional. At $z = 0$, clarity is 0.00 (maximally ambiguous) and the briefing presents a balanced picture.

Table A3: Identification map. Each row lists a candidate channel through which the surveillance treatment could produce the observed action effect; the right column identifies the test(s) addressing it and the conclusion. "Inconclusive" marks channels left bundled by the current design.

Candidate channel	Test & outcome
Direct deterrence at decision time	Decision prompt held fixed in main treatment (no monitoring reference; replication-package configuration flag `decision_context=none`). A separate hold-fixed test (§5) shows this channel *exists* when the warning is moved to the decision prompt ($\Delta = -20.9$ pp), but it is absent in the main design. *Ruled out as the main-treatment driver.*
Generic observation effect	Placebo ("monitoring for research, no consequences") and anonymous monitoring variants (Appendix E.4). Both are null in the primary model ($p > 0.9$); Llama variants are small but imprecise. *Disfavored as the main driver.*
Generic message-stage degradation	Codedness-induction control (§5, Llama 3.3 70B, 500 nested cells): maximally coded messages without monitoring framing move joining by only -0.7 pp, against -8.0 pp under surveillance on the same cells. Cross-task discriminant test: surveillance moves the private-bet task by 5.6 pp, but the bet can still embed participation expectations. *Generic style degradation ruled out as the main driver.*
Coordination-cue loss	Full prompt-isolation coord-task delta compared with cross-task delta (Llama: 15.0 vs. 5.6 pp). The samples are not cell-matched; the rotation-matched Llama subset is smaller and closer to the private-bet estimate; and the private bet can still embed participation expectations. This provides *descriptive mechanism evidence* rather than a point-identified residual.
Risk/fear salience in messages	Punishment-risk elicitation across conditions (Table A17): ratings $\approx 8.0/10$ everywhere, possibly ceiling-limited. Receiver-rationale audit (§5): fear/caution references *fall* under surveillance (84.2% $\rightarrow$ 69.8%) and explicit monitoring references are rare in both arms. *Disfavored as the dominant receiver-side channel, though the dictionary evidence is descriptive and tone is not separately randomized.*
Model-style reflexivity	Bidirectional cross-model rotation on matched cells. *Llama writes $\rightarrow$ Qwen3 30B reads*: -7.5 pp ($N = 200$, $p < 0.001$), comparable to within-Llama matched (-6.0 pp). *Qwen3 30B writes $\rightarrow$ Llama reads*: -1.9 pp ($N = 200$, $p < 0.001$). Both significant. *Ruled out as the driver*, with cross-direction asymmetry noted.

Table A4: Classifier baselines. Accuracy and AUC are 5-fold CV on pure-treatment data. "Pred. join (surv.)" is the classifier's predicted join rate when applied to surveillance-treatment briefings; "Actual" is the LLM's observed rate. The gap measures the surveillance action shift invisible to briefing-body text classifiers.

Classifier	Acc.	AUC	Pred. (surv.)	Actual	Gap
BoW TF-IDF	88.5%	0.947	40.6%	27.6%	+13.0 pp
Slider logistic	88.4%	0.941	40.4%	27.6%	+12.8 pp
Keyphrase	60.2%	0.594	8.4%	27.6%	−19.2 pp

Notes: Primary model (Mistral Small Creative). Accuracy and AUC from 5-fold cross-validation on pure-treatment data. Training sample: the primary model's pure arm (1,000 country–period rows, 25 agents per row, 25,000 agent decisions); each CV fold trains on 80% and tests on the held-out 20%. The surveillance application scores all 1,000 surveillance-arm rows (25,000 agent decisions). Gap = classifier-predicted join rate − actual LLM join rate under surveillance (pp).

Table A5: Narrative-header conditions: classifier vs. actual LLM behavior. Actual join rates shift sharply across strategic-stakes framing conditions; the classifier, trained on briefing-body text features alone, cannot predict this header-induced shift.

Condition	N	Classif. pred.	Actual	Gap (pp)
Baseline (no header)	270	40.9%	40.9%	+0.0
High-cost header	270	40.9%	19.0%	+21.9
Low-cost header	270	40.9%	69.3%	−28.4

Notes: Primary model (Mistral Small Creative). N counts country–period rows on the information-design θ-grid ($n = 25$ agents per row). Logistic classifier trained on baseline slider features from the briefing body. Gap = classifier-predicted join rate − actual LLM join rate (pp). The sign convention is the reverse of Table A19, which reports Δ = actual − baseline: because the classifier's prediction stays near the baseline rate, a header that lowers actual joining shows up there as a negative Δ and here as a positive Gap of approximately the same magnitude. High/low rows are the two narrative-header conditions on the information-design grid, not numeric payoff-ladder endpoints.

Table A6: Top 10 most discriminative uni- and bi-grams between baseline and surveillance peer messages (primary model). A logistic classifier on tf-idf features achieves 91.9% accuracy on a held-out sample of $N = 15{,}000$.

Surveillance-distinguishing		Baseline-distinguishing	
Feature	Coef	Feature	Coef
`things feel`	+6.41	`look the`	−11.11
`walls are`	+5.87	`look`	−8.93
`are cracking`	+5.64	`listen`	−7.31
`things`	+5.56	`listen this`	−6.00
`cracking even`	+4.96	`may`	−4.93
`cracking`	+4.46	`isn just`	−4.68
`ground shifting`	+3.79	`don panic`	−4.63
`the walls`	+3.54	`the weather`	−4.49
`feel like`	+3.50	`weather`	−4.38
`rush this`	+3.37	`something shifting`	−4.26

Notes: Logistic regression on tf-idf uni- and bi-grams (max 5,000 features, min document frequency 10). Train/test split 70/30, stratified ($N_{\text{train}} = 35{,}000$, $N_{\text{test}} = 15{,}000$). Positive coefficients distinguish surveillance messages, negative coefficients distinguish baseline. The sample contains only baseline communication and clean surveillance peer messages; placebo/weather-stakes conditions are excluded. The weather terms are organic peer-message metaphors, not placebo leakage: 2,268 baseline messages (9.07%) and 123 surveillance messages (0.49%) contain the token `weather`. Direct-reference and coded-metaphor claims in the text come from separate dictionary counts, not from this top-n-gram list alone.

Table A7: Communication estimators: aggregation accounting.

Model	Rows/arm	Row wt. (P/C)	Cells/arm	Matched	Cell wt.	Lost (P/C)	Δ unpaired (pp)	Δ paired (pp)
Mistral Small Creative	1000	55.6%/38.5%	680	680	46.0%	0/0	+2.38	+4.84
Llama 3.3 70B	100	5.6%/3.8%	100	100	6.8%	0/0	−2.36	−2.36
Qwen3 30B	100	5.6%/3.8%	100	100	6.8%	0/0	−4.64	−4.64
GPT-OSS 120B	200	11.1%/7.7%	200	200	13.5%	0/0	+3.47	+3.47
Qwen3 235B	200/1000	11.1%/38.5%	200/1000	200	13.5%	0/800	+0.86	+0.12
Trinity Large	100/99	5.6%/3.8%	100/99	99	6.7%	1/0	−3.14	−3.41
MiniMax M2-Her	100	5.6%/3.8%	100	100	6.8%	0/0	+1.32	+1.32
Equal-weight avg	—	14.3% each	—	—	14.3% each	—	−0.30	−0.09
Pooled	1800/2599	100.0%	1480/2279	1479	100.0%	1/800	+1.39	+2.10

Notes: Sample: valid country–period rows from the core pure and communication suites, each aggregating $n = 25$ agent decisions. "Rows/arm" counts valid country-period rows entering the pooled unpaired estimator in each arm. "Cells/arm" counts unique task cells after collapsing duplicates by the exact matching key (model, country, period, θ, z, benefit, θ^*). The paired estimator averages within these cells and keeps only common support; "Row wt." and "Cell wt." are the model shares in the pooled unpaired and pooled paired estimators, respectively. The pooled paired Δ of +2.10 pp has a paired standard error of 0.33 pp ($t = 6.38$, $p < 0.001$, $N = 1{,}479$ matched cells). Trinity Large has one pure-only cell. Qwen3 235B has 800 communication-only cells because its communication arm is larger than its pure arm; those cells are excluded from the paired common-support estimator.

Table A8: Temperature robustness across three models. The pure global game is run at varying LLM decoding temperatures. The correlation $r(J, A(\theta))$ is stable across all temperatures and models.

Model	T	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$
Mistral	T=0.3	100	41.2%	+0.83	−0.05
Mistral	T=0.7	100	40.6%	+0.82	−0.08
Mistral	T=1.0	100	41.0%	+0.86	−0.04
Llama 3.3 70B	T=0.3	100	36.4%	+0.78	−0.21
Llama 3.3 70B	T=0.5	100	36.0%	+0.78	−0.22
Llama 3.3 70B	T=0.7	100	36.1%	+0.78	−0.22
Llama 3.3 70B	T=1.0	100	36.0%	+0.78	−0.22
Llama 3.3 70B	T=1.2	100	35.8%	+0.78	−0.23
Qwen3 235B	T=0.3	100	39.2%	+0.83	−0.12
Qwen3 235B	T=0.5	100	39.2%	+0.82	−0.11
Qwen3 235B	T=0.7	100	39.2%	+0.84	−0.11
Qwen3 235B	T=1.0	100	39.8%	+0.83	−0.11
Qwen3 235B	T=1.2	100	39.7%	+0.83	−0.12

Notes: Pure treatment, varying LLM decoding temperature. N counts country–period rows ($n = 25$ agents per row); mean join in percent. Mistral was run at only three temperatures (0.3, 0.7, 1.0) because the hosted endpoint was retired before the 0.5 and 1.2 cells could be collected; Llama 3.3 70B and Qwen3 235B cover the full five-point ladder.

Table A9: Clean prompt-isolation surveillance reruns. The surveillance prompt is the baseline trusted-contact communication prompt plus one appended monitoring warning; the decision prompt is unchanged.

Model	Full cells (B/S)	Matched cells	Baseline	Surv.	Δ (pp)	SE	$r(J, A)$	p
Mistral Small Creative	680/1000	500	45.3%	31.4%	−13.9	0.7	+0.72	<0.001
Llama 3.3 70B	100/1000	20	46.2%	31.2%	−15.0	4.4	+0.65	0.003
Qwen3 30B	100/1000	20	50.0%	37.8%	−12.2	2.5	+0.68	<0.001
GPT-OSS 120B	200/1000	20	42.5%	30.5%	−12.0	3.9	+0.75	0.006
Qwen3 235B	1000/1000	20	42.0%	39.0%	−3.0	2.2	+0.78	0.186
Equal-weight avg	—	—	—	—	−11.2	—	—	—
Pooled matched	—	580	—	—	−13.5	0.7	—	<0.001

Notes: Baseline is each model's communication treatment. "Full cells" counts the available baseline and surveillance task cells after collapsing duplicate rows by the exact key (country, period, θ, z, benefit, θ^*); it is not the support used for the paired effect when common support is smaller. The non-primary baseline communication arms were shorter pilot grids (100–200 cells) while the later surveillance reruns used a larger grid; because those grids were not nested, only 20 exact cells overlap for several models. N for each paired contrast is the number of matched common-support cells; Baseline, Surv., Δ, SE, and p are computed only on those cells, with p from a paired one-sample t-test of cell-level (surveillance − baseline) differences and SE the paired standard error of Δ in pp. $r(J, A)$ is the surveillance-arm threshold-alignment correlation on the full surveillance rerun. Aggregating the primary model's matched-cell deltas to country means ($N = 5$ countries) gives $t = -13.90$, $p < 0.001$, so the headline effect survives country-level clustering.

Table A10: Surveillance isolation checks. Placebo: monitored for research, no consequences. Anonymous: messages aggregated anonymously. Neither deviates significantly from the communication baseline.

Model	Variant	N	Mean join	$r(J, A)$	Δ (pp)	p
Mistral	Placebo	200	41.4%	+0.73	+0.3	0.935
	Anonymous	200	41.2%	+0.73	+0.1	0.979
Llama 3.3 70B	Placebo	1000	41.2%	+0.71	−0.3	0.940
	Anonymous	1000	40.5%	+0.72	−1.1	0.814

Notes: Placebo: agents told monitoring is for research purposes only (no consequences). Anonymous: messages aggregated before delivery. N counts country–period rows ($n = 25$ agents per row). Δ: change vs. the same model's communication baseline (pp), with p from a two-sample t-test against that baseline. Communication-baseline arms: Mistral: $N = 1000$, mean join 41.1%; Llama 3.3 70B: $N = 100$, mean join 41.6%.

Table A11: Every Llama 3.3 70B surveillance estimate: effect size, matched support, grid, and purpose. All deltas are surveillance (or the stated arm) minus its own baseline on the matched cells. The full-scale nested estimate (−8.0 pp, $N = 500$) is the headline cross-model confirmation; the original prompt-isolation estimate (−15.0 pp) rests on only 20 overlapping cells and is a noisy sign check, not a full replication.

Arm	Δ (pp)	Matched N	Grid	Purpose
Surveillance (monitoring warning), coordination task:				
Nested clean surveillance	−8.0	500	nested 10×50, common seed	full-scale confirmation (§5)
Prompt-isolation (original)	−15.0	20	20-cell pilot/rerun overlap	original sign check (Table A9)
Mild routine-monitoring note	−6.8	500	nested 10×50	dose response (Appendix E.5)
Severe-consequence note	−10.0	500	nested 10×50	dose response (Appendix E.5)
Replay / decoding of surveilled messages, coordination task:				
Raw-message replay	−7.9	500	nested 10×50	validates replay design (≈ live)
Decoded-message replay	−4.9	500	nested 10×50	decoding restores participation
Cross-model writer–reader rotation:				
Within-Llama (matched baseline)	−6.0	200	rotation cells	reflexivity baseline (§5)
Llama writes, Qwen3 30B reads	−7.5	200	rotation cells	cross-family transmission
Qwen3 30B writes, Llama reads	−1.9	200	rotation cells	cross-family transmission
Other decision task:				
Private bet on regime fall	−5.6	500	cross-task grid	task contrast (§5)
Reference control (no monitoring framing):				
Codedness-induction control	−0.7	500	nested 10×50	isolates coded form alone

Notes: "Matched N" is the number of common-support cells entering each paired contrast. "Nested" arms share the identical 500-cell grid (10 countries $\times 50$ periods, $n = 25$, common seed) and are matched by construction. The original prompt-isolation estimate shares only 20 cells with its pilot baseline because those grids were not nested. Rotation contrasts are matched on (country, period, θ) cells across writer and reader models. The private-bet and nested-coordination samples are not cell-matched to each other.

Table A12: Finite-n benchmark: predicted vs. empirical regime fall rates

Model	N cells	Logistic x_0	Pearson r	RMSE	MAE
Mistral	600	−0.11	0.997***	0.042	0.015
Llama 3.3 70B	100	0.04	0.957***	0.148	0.053
Qwen3 30B	100	0.20	0.987***	0.083	0.026
GPT-OSS 120B	200	−0.01	0.997***	0.038	0.014
Qwen3 235B	200	−0.04	0.988***	0.076	0.025
Trinity	100	0.31	0.980***	0.102	0.039
MiniMax M2	100	1.23	0.999***	0.026	0.012
Pooled	1400	−0.05	0.999***	0.019	0.007

Notes: Pure treatment only, $n = 25$ agents per row. For each θ bin, the predicted fall rate is $\Pr(\text{Binom}(n, \hat{p}(\theta)) > n\theta)$ where $\hat{p}(\theta)$ is the fitted logistic join probability. "N cells" is the number of unique country–period cells after pooling repeated log entries; for Mistral the pure-treatment log contains 1,000 runs over 600 unique (country, period) indices (re-runs appended to the same log are pooled within a cell). Because θ is re-drawn whenever a batch is appended, those 600 (country, period) indices span 680 unique (country, period, θ) draws—the matching key on which the communication and surveillance estimators operate (Table A7); this finite-n check instead pools all θ draws within a (country, period). "Logistic x_0" is the midpoint of the logistic $\hat{p}(\theta) = L/(1 + e^{-k(\theta - x_0)})$ fitted to bin-averaged join rates on a random 70% train split of cells (fixed seed); it can therefore differ from the full-sample cutoff in the logistic-parameters table, which is estimated on all observations without the split. Pearson r is the correlation between predicted and empirical fall rates across θ bins (10–15 bins per model, 15 pooled), computed in-sample over all cells; stars are from the two-sided test of zero correlation across those bins. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A13: Agent-level regressions

	(1) Join Decision Logit		(2) Coord. Ablation Logit		(3) Belief $\rightarrow$ Action Logit	
θ	-1.745^{***}	(0.043)				
Direction			-14.088^{***}	(0.996)		
Coordination			-9.333^{***}	(1.131)		
Dir $\times$ Coord			-0.042	(0.360)		
Belief					0.353^{***}	(0.023)
z-score					-0.026	(0.082)
Comm	0.017	(0.023)				
Flip	-0.007	(0.055)				
Scramble	0.068^{*}	(0.038)				
Surveillance	-1.596^{***}	(0.048)				
$\theta\times$ Comm	-0.538^{***}	(0.037)				
$\theta\times$ Flip	3.308^{***}	(0.070)				
$\theta\times$ Scramble	1.702^{***}	(0.044)				
$\theta\times$ Surveillance	-1.151^{***}	(0.062)				
Treat: Pure					4.673^{***}	(0.981)
Treat: Surveillance					0.021	(0.244)
Constant	-0.086^{*}	(0.051)	11.022^{***}	(1.065)	-25.429^{***}	(1.686)
Model FE	Yes		No		No	
Clustered SE	Yes		Yes		Yes	
N	240,557		44,662		14,990	
Pseudo R^2	0.353		0.442		0.975	

Notes: Logit coefficients with standard errors clustered at the model–country–period level in parentheses (all three columns; in column (3) the sample is almost entirely a single model, so clusters effectively coincide with country–period cells). Column (1): agent-level join decisions pooled across all seven models in the pure, communication, scramble, and flip treatments, plus surveillance runs for Mistral, Llama 3.3 70B, and Qwen3 30B ($N = 240{,}557$; 4,700 clusters). Regressors: θ, treatment dummies (base category: pure), $\theta\times$ treatment interactions, and model fixed effects. Column (2): coordination ablation—$\Pr(\text{join}_i = 1) = \Lambda(\beta_0 + \beta_1 \text{Dir}_i + \beta_2 \text{Coord}_i + \beta_3 \text{Dir}_i \times \text{Coord}_i)$, where Λ denotes the logistic CDF—using briefing slider values; pure treatment only, all seven models pooled, no model fixed effects ($N = 44{,}662$; 1,400 clusters). Direction $\in [0, 1]$ is increasing in regime strength; Coordination $\in [0, 1]$ is decreasing (high = weaker regime). Because both are monotone logistic transforms of the same signal, they are near-collinear ($\rho \approx -1$); individual coefficients reflect the partial effect *conditional on the other*, not the marginal effect. Column (3): partial effect of elicited belief ($b_i \in [0, 100]$, stated success probability) on action, controlling for z-score. Sample: belief-elicitation runs only (communication, pure, and surveillance variants), comprising Mistral (14,981 obs.), Llama 3.3 70B (9 obs.); $N = 14{,}990$; 369 clusters. "Treat: X" rows are treatment intercept dummies (base category: communication), not interactions with belief. Marginal effects, column (1): for the continuous regressor θ, the derivative-based effect at the mean is $\hat{\beta}_\theta\, \bar{p}(1 - \bar{p}) = -0.416$. For the binary surveillance treatment, a derivative-based marginal effect is not meaningful and would ignore the $\theta\times$ surveillance interaction; we therefore report the average discrete-change effect of surveillance (toggling the dummy and its θ-interaction jointly, with all other regressors at observed values, averaged over the column (1) estimation sample): -17.9 pp. **Caveat:** beliefs are elicited *after* the join/stay decision (post-decision), so the belief coefficient reflects association rather than a causal pathway; post-decision rationalization may inflate the correlation. The very high pseudo-R^2 (0.975) and large intercept indicate near-deterministic prediction consistent with quasi-separation—belief almost perfectly predicts action, as expected when belief is stated *post hoc*. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A14: Logistic fit parameters by model and treatment. $\hat{\theta}^*$ is the estimated cutoff ($-b_0/b_1$); $\kappa \equiv b_1$ is the steepness parameter.

	Pure			Communication		
Model	$\hat{\theta}^*$ (SE)	κ (SE)	N	$\hat{\theta}^*$ (SE)	κ (SE)	N
Mistral Small Creative	−0.32 (0.02)	+2.15 (0.08)	1000	−0.23 (0.02)	+2.57 (0.11)	1000
Llama 3.3 70B	+0.02 (0.04)	+2.83 (0.36)	100	−0.06 (0.04)	+3.63 (0.52)	100
Qwen3 30B	+0.10 (0.06)	+1.62 (0.16)	100	−0.03 (0.04)	+2.94 (0.36)	100
GPT-OSS 120B	−0.25 (0.04)	+2.06 (0.15)	200	−0.15 (0.03)	+3.03 (0.26)	200
Qwen3 235B	−0.22 (0.03)	+2.11 (0.15)	200	−0.18 (0.01)	+2.78 (0.10)	1000
Trinity Large	+0.08 (0.05)	+1.34 (0.11)	100	−0.02 (0.04)	+2.23 (0.23)	100
MiniMax M2-Her	−0.17 (0.09)	+0.61 (0.06)	100	−0.07 (0.09)	+0.66 (0.06)	100

Notes: Fitted form: $P(\text{join} \mid \theta) = 1/(1 + e^{b_0 + b_1\theta})$. $\hat{\theta}^* = -b_0/b_1$: estimated cutoff. $\kappa \equiv b_1$: logistic steepness (larger = sharper threshold). Sign convention: positive κ corresponds to a join curve decreasing in θ; in the agent-level logit (Table A13) the same comparative static appears as a negative coefficient on θ. N counts period-level rows entering each fit. Standard errors from the covariance matrix of the nonlinear fit; cutoff SE by delta method.

Table A15: Cross-generator robustness. Three text rendering styles (baseline, diplomatic cable, journalistic wire) use identical slider functions and evidence items; only prose formatting differs. The Pearson $r(J, A(\theta))$ and logistic cutoff are virtually identical across generators.

Model	Generator	N	Mean join	$r(J, A)$ [95% CI]	$\hat{\theta}^*$
Mistral	Baseline	100	40.2%	+0.83 $[0.76, 0.88]$	−0.07
Mistral	Cable	100	40.6%	+0.83 $[0.75, 0.88]$	−0.06
Mistral	Journalistic	100	40.3%	+0.83 $[0.76, 0.88]$	−0.06
Llama 3.3 70B	Baseline	100	35.4%	+0.77 $[0.67, 0.84]$	−0.25
Llama 3.3 70B	Cable	100	36.2%	+0.77 $[0.68, 0.84]$	−0.23
Llama 3.3 70B	Journalistic	100	36.0%	+0.78 $[0.68, 0.84]$	−0.23

Notes: N counts country–period rows ($n = 25$ agents per row); mean join in percent. 95% Fisher-z confidence intervals for r in brackets. Cutoff $\hat{\theta}^*$ from the same logistic specification as Table A14, reported to two decimals. Runs use the core settings ($\sigma = 0.3$, temperature 0.7).

Table A16: Agent-level belief correlations (primary model: Mistral Small Creative).

	N	$r_{\text{belief,posterior}}$	$r_{\text{belief,decision}}$	r_{partial}	Mean belief (%)
Pure	10,000	+0.80 $[0.79, 0.80]$	+0.84 $[0.83, 0.85]$	+0.64	44.4
Comm.	5,000	+0.80 $[0.79, 0.81]$	+0.83 $[0.82, 0.84]$	+0.56	44.4
Surv.	5,000	+0.79 $[0.78, 0.80]$	+0.73 $[0.72, 0.74]$	+0.47	43.6

Notes: N counts agent-level observations (one elicited belief per agent decision). Beliefs are elicited on a 0–100 probability scale; mean belief is reported on that percent scale. $r_{\text{belief,posterior}}$: stated belief vs. the theoretical posterior $P(\theta < \theta^* \mid x_i)$ implied by the agent's signal. $r_{\text{belief,decision}}$: stated belief vs. the JOIN/STAY decision. r_{partial}: belief–decision partial correlation controlling for the signal z-score. 95% Fisher-z confidence intervals in brackets.

Table A17: Elicited punishment risk (0–10 scale). Agents rate expected regime punishment after their JOIN/STAY decision. "JOIN" and "STAY" columns show the mean rating conditional on the agent's own decision.

Model	Cond.	N	Mean (SD)	| JOIN	| STAY
Mistral	Pure	2,500	8.1 (0.9)	8.0	8.2
	Comm.	2,500	8.1 (0.9)	8.0	8.1
	Surv.	2,500	8.1 (0.9)	8.1	8.1
Llama 3.3 70B	Pure	2,500	8.0 (0.2)	8.0	7.9
	Comm.	2,500	7.9 (0.2)	8.0	7.9
	Surv.	2,500	7.9 (0.2)	8.0	7.9

Notes: Agent-level elicitation on a 0–10 scale; N counts agent-level ratings (one per agent decision). Standard deviations of the overall rating in parentheses; conditional means are reported without dispersion measures, which are not stored in the verified statistics. The punishment-risk module was fielded only on Mistral and Llama 3.3 70B; the other five models were not run with this elicitation.

Table A18: Stakes conditions. *Top:* B/C payoff/signal-grid sweep, in which (B, C) vary and the θ and z-score support shift with the target θ^*. *Bottom:* narrative stakes ladder, in which only a prepended header varies.

Condition / target θ^*	B	C	Header / narrative framing
Payoff sweep (standard briefings, no header):			
$\theta^* = 0.25$	0.33	1	—
$\theta^* = 0.33$	0.49	1	—
$\theta^* = 0.45$	0.82	1	—
$\theta^* = 0.50$	1.00	1	—
$\theta^* = 0.60$	1.50	1	—
$\theta^* = 0.67$	2.03	1	—
$\theta^* = 0.75$	3.00	1	—
Narrative cost ladder (B=C=1; header prepended):			
Very high cost	1	1	Disappearances, families targeted
Moderate high	1	1	Prison sentences, job loss
Neutral	1	1	Consequences uncertain
Moderate low	1	1	Brief questioning, released
Very low cost	1	1	Guaranteed asylum, zero cost
Controls:			
Placebo	1	1	Weather information (irrelevant)
High cost v2	1	1	Paraphrase of very high cost
Low cost v2	1	1	Paraphrase of very low cost

Notes: Payoff sweep: $B = \theta^*/(1-\theta^*)$, C=1. The z-score support and θ-grid shift with θ^*, so the briefing-text distribution is approximately constant across the seven cells. Narrative ladder: B=C=1 in every row; only the prepended header varies. Join rates: very high cost 21.3%, moderate high 46.6%, neutral 46.7%, moderate low 68.4%, very low 78.9%. Placebo 32.7%, v2 variants 27.4%/63.8%. Full headers in Appendix A.

Table A19: Strategic-stakes narrative-header comparative statics. High-cost framing emphasizes severe reprisals for failed action; low-cost framing emphasizes minimal consequences. These rows are the two-header narrative contrast, not the seven-cell numeric payoff ladder.

Design	N	Mean join	$r(J, A)$	$\hat{\theta}^*$ (SE)	Δ (pp)
Baseline (no header)	270	40.9%	+0.88	+0.39 (0.007)	—
High-cost header	270	19.0%	+0.87	+0.13 (0.010)	−21.9
Low-cost header	270	69.3%	+0.88	+0.72 (0.007)	+28.4

Notes: Primary model, information-design θ-grid. N counts country–period rows ($n = 25$ agents per row). Identical briefing bodies across conditions; only the prepended strategic-stakes header varies. The high/low rows are the original header conditions `bc_high_cost` and `bc_low_cost`; they are not the endpoints of the separate narrative ladder reported in Table A18. Δ: change in mean join vs. baseline (pp, actual − baseline); cutoff SEs are delta-method standard errors from the logistic fit (per-condition mean-join SEs are not stored in the verified statistics). Note the sign convention relative to Table A5: there, Gap = classifier − actual, and because the classifier's prediction stays near the baseline rate, a header that lowers actual joining appears here as a negative Δ and there as a positive Gap of approximately the same magnitude.

Table A20: Parse error and API failure rates by model and treatment.

Model	Treat.	N	API err	Unparse.	Combined
Mistral	Pure	1000	0.0%	0.0%	0.0%
	Comm	1000	0.0%	0.0%	0.0%
	Scr.	200	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Llama 3.3 70B	Pure	100	0.0%	0.0%	0.0%
	Comm	100	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Qwen3 30B	Pure	100	0.0%	0.0%	0.0%
	Comm	100	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
GPT-OSS 120B	Pure	200	0.0%	1.5%	1.5%
	Comm	200	0.0%	0.3%	0.3%
	Scr.	500	0.0%	3.1%	3.1%
	Flip	500	0.0%	3.3%	3.3%
Qwen3 235B	Pure	200	0.0%	0.0%	0.0%
	Comm	1000	0.0%	0.0%	0.0%
	Scr.	100	0.0%	0.0%	0.0%
	Flip	100	0.0%	0.0%	0.0%
Trinity	Pure	100	9.0%	0.0%	9.0%
	Comm	100	10.0%	0.0%	10.0%
	Scr.	100	8.0%	0.0%	8.0%
	Flip	100	9.9%	0.0%	9.9%
MiniMax M2	Pure	100	0.0%	1.5%	1.5%
	Comm	100	0.0%	0.9%	0.9%
	Scr.	100	0.0%	1.8%	1.8%
	Flip	100	0.0%	1.0%	1.0%

Notes: Per-period averages. API error = provider-side failure (timeout, rate limit, content filter). Unparseable = valid response not classified as JOIN/STAY. Combined $< 2\%$ in every listed treatment for five of seven models. The two exceptions are GPT-OSS 120B, with 3.1–3.3% unparseable responses in the scramble and flip arms, and Trinity, with 8–10% API errors from provider content filtering. Trinity's elevated API errors ($\approx 9\%$) mean its missingness is reported rather than resolved. A signal-balance diagnostic for Trinity's per-period API-error rate gives max $|r(\theta, \text{error})| = 0.19$ over the listed treatments (minimum $p = 0.064$).

Table A21: Slider values at representative z-scores.

z	Direction	Clarity	Coordination
−2.0	0.17	0.98	0.77
−1.0	0.31	0.63	0.65
0.0	0.50	0.00	0.50
+1.0	0.69	0.63	0.35
+2.0	0.83	0.98	0.23